HSPT Exam Simulator

Full-Length Practice Exam 16

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. After missing lunch and walking all afternoon, Niko was RAVENOUS. RAVENOUS most nearly means:

A) slightly impatient

B) extremely hungry

C) unusually sleepy

D) pleasantly surprised

2. The abandoned shed was DILAPIDATED, with a sagging roof and broken boards. DILAPIDATED most nearly means:

A) in a state of serious disrepair

B) recently enlarged

C) carefully concealed

D) unusually expensive

3. The historian made a FERVENT appeal to preserve the recordings. FERVENT most nearly means:

A) intensely passionate

B) deliberately confusing

C) barely audible

D) strictly impartial

4. Which word is most nearly opposite to SLUGGISH when describing a worker's movements?

A) lethargic

B) slow

C) torpid

D) brisk

5. Attendance at the briefing is OBLIGATORY. Which word most nearly expresses the opposite of OBLIGATORY?

A) mandatory

B) compulsory

C) optional

D) required

6. Which word is most nearly opposite to MOROSE, describing someone's mood?

A) cheerful

B) sullen

C) gloomy

D) doleful

7. Changing a length from centimeters to meters changes its expression without changing the length itself. A faithful translation similarly:

A) adds a new claim that the original never made

B) replaces the original subject with an unrelated subject

C) removes every detail needed to understand the original

D) changes the language while preserving the stated meaning

8. A sieve lets smaller particles pass while holding larger ones back. Which action follows a similar selective principle?

A) A program deletes every incoming message without examining it.

B) A printer makes a second copy of every message.

C) An email filter separates marked junk messages while admitting other messages.

D) A program rearranges all messages alphabetically without excluding any.

9. A terrain model exaggerates heights while keeping horizontal positions unchanged, making slopes easier to notice. Which operation changes another dimension in a comparable way?

A) A slow-motion replay expands time intervals while preserving the movement's path and order.

B) A reversed replay makes events occur in the opposite order.

C) A cropped photograph removes everything near its edges.

D) An invented map adds a mountain that does not exist.

10. Which description names a span of time rather than a single point in time?

A) Noon

B) The instant of arrival

C) An intermission

D) The scheduled starting moment

11. Which group contains ONLY changes between physical states of matter?

A) boiling; walking; melting

B) melting; freezing; evaporating

C) drifting; freezing; floating

D) climbing; turning; sinking

12. Which pair names complementary roles that require each other by definition, rather than roles that are only sometimes associated?

A) writer; publisher

B) singer; conductor

C) lender; borrower

D) traveler; guide

13. Three notice boards display the same weather report, but all copied it from one original notice. A reader calls this three independent confirmations. Why is that reasoning weak?

A) A report becomes false whenever someone copies it.

B) Weather reports can be confirmed only by being printed in color.

C) Different boards cannot display identical wording.

D) The agreement comes from a shared source rather than separate checks.

14. It is false that every repaired instrument that passed inspection was delivered. Which statement must be true?

A) Every repaired instrument was withheld from delivery.

B) At least one repaired instrument passed inspection but was not delivered.

C) At least one undelivered instrument failed inspection.

D) Every delivered instrument failed inspection.

15. Three switches are each on or off. P is on exactly when Q is off. Q is on exactly when R is on. R is on exactly when P is on. Which conclusion is correct?

A) All three switches can be on while satisfying every rule.

B) The three rules cannot all be satisfied.

C) P and R must have opposite states under the rules.

D) All three switches can be off while satisfying every rule.

16. The instructions already included a clear diagram, so a second identical diagram was SUPERFLUOUS. SUPERFLUOUS most nearly means:

A) misleading

B) incomplete

C) unnecessary

D) compulsory

17. The stain was INDELIBLE; repeated washing could not remove it. INDELIBLE most nearly means:

A) impossible to erase

B) difficult to notice

C) quick to spread

D) pleasant to recall

18. A RANCOROUS argument left both neighbors nursing bitter resentment. RANCOROUS most nearly means:

A) conducted in secret

B) settled without delay

C) based on careful measurements

D) filled with deep hostility

19. The hillside had only SPARSE vegetation. Which word most nearly expresses the opposite of SPARSE here?

A) dense

B) scattered

C) meager

D) thin

20. The arrangement was declared IRREVOCABLE. Which word most nearly expresses the opposite of IRREVOCABLE?

A) binding

B) revocable

C) unalterable

D) permanent

21. The hall had a STAID atmosphere, restrained and solemn. Which word most nearly expresses the opposite of STAID?

A) sedate

B) dignified

C) sober

D) exuberant

22. A photograph negative reverses light and dark while keeping the positions of features. Which operation preserves positions while similarly reversing a pair of opposed labels?

A) Moving the last checklist entry to the beginning

B) Erasing entries that seem unimportant

C) Changing every yes on a fixed checklist to no and every no to yes

D) Making an additional identical copy of the checklist

23. A bookmark records where a reader stopped so reading can resume there. Which feature serves the most similar purpose?

A) A cover illustration that suggests a story's subject

B) A saved checkpoint that lets a game resume from recorded progress

C) A clock that reports the present time

D) A list that ranks games by popularity

24. A removable casting model is sacrificed to create a cavity that the finished material will occupy. Which action follows the most similar relationship?

A) Using a temporary spacer, then removing it so a permanent piece can fit in the reserved opening

B) Adding a decorative border that remains around the finished piece

C) Measuring a finished piece after all construction is complete

D) Polishing a surface without changing which parts occupy it

25. Which document records that payment has been received rather than proposing a price before a purchase or job?

A) estimate

B) receipt

C) quotation

D) bid

26. Which group consists entirely of words that can describe a result produced without deliberate intention?

A) accidental; calculated; unplanned

B) deliberate; inadvertent; designed

C) accidental; inadvertent; unplanned

D) planned; calculated; intentional

27. Which word does NOT belong with HUSK, RIND, and PEEL as an outer covering that can be removed from food?

A) shell

B) skin

C) kernel

D) hull

28. A hinged door has an accurate PUSH sign on one face and an accurate PULL sign on its opposite face. Why need the signs not contradict each other?

A) Both signs require one person to push and pull at the same moment.

B) Opposite verbs necessarily make one of the signs inaccurate.

C) Both signs prohibit anyone from moving the door.

D) Actions taken from opposite sides can move the same door in the same direction.

29. An inspection rule says a crate may receive a seal only if both its lock and its lining pass inspection. A sealed crate is found with a lining that failed. Which statement must be true?

A) The lock must also have failed.

B) The sealing rule was violated, regardless of the lock's result.

C) The lining must have passed because a seal is present.

D) The rule permits a seal whenever the lock passes.

30. Ravi says, "The engine is idling or the indicator is lit (or both)." Lena says, "If the engine is idling, the indicator is lit." Both statements are true. What must be true?

A) The engine is idling.

B) The engine is not idling.

C) The indicator is not lit.

D) The indicator is lit.

31. Lena grew PENSIVE as she considered the question in silence. PENSIVE most nearly means:

A) openly boastful

B) physically restless

C) deeply thoughtful

D) eager to interrupt

32. The cottage's tiny windows and old-fashioned shutters gave it a QUAINT appearance. QUAINT most nearly means:

A) dangerously unstable

B) strictly modern

C) charmingly old-fashioned

D) deliberately hidden

33. The review's CAUSTIC remarks wounded the performer's pride. CAUSTIC most nearly means:

A) mildly encouraging

B) harshly biting

C) carefully numerical

D) deliberately vague

34. A FUTILE effort achieves nothing useful. Which word most nearly expresses the opposite of FUTILE?

A) pointless

B) useless

C) unproductive

D) effective

35. The agreement was CLANDESTINE, known only to its secret participants. Which word most nearly expresses the opposite of CLANDESTINE?

A) public

B) covert

C) concealed

D) undercover

36. The critic called the description PROSAIC, ordinary and uninspired. Which word most nearly expresses the opposite of PROSAIC here?

A) imaginative

B) mundane

C) commonplace

D) unremarkable

37. A Braille cell distinguishes symbols through the positions of raised dots. Which feature uses position to distinguish another kind of information?

A) A metronome produces regularly timed clicks.

B) A conductor signals when a group should begin.

C) A padded case protects an instrument during travel.

D) The height of a note on a musical staff indicates its pitch.

38. A thermostat changes a heater's operation in response to measured temperature. Which action uses a comparable feedback process?

A) A potter examines each fired test tile and adjusts the next firing according to the result.

B) A potter follows an unchanged firing schedule without checking any results.

C) A clerk records firing dates but never changes the procedure.

D) A customer selects a finished tile solely because its color matches a wall.

39. A subscription's expiration ends access unless the subscriber renews it. Which relationship follows the same time-limited authorization pattern?

A) A receipt records that a payment already occurred.

B) A temporary pass stops permitting entry when its stated period ends unless it is extended.

C) A biography describes events that happened long ago.

D) A permanent scratch remains visible after an object is moved.

40. Which term names an animal's den rather than a type of manufactured container?

A) crate

B) canister

C) carton

D) lair

41. Which group contains ONLY words that name an action that, by definition, undoes a corresponding earlier action?

A) untie; polish; enlarge

B) untie; disconnect; unwrap

C) decorate; disconnect; photograph

D) unwrap; inspect; measure

42. Which expression does NOT belong with the others as a way of conveying ridicule?

A) a sneering imitation

B) a sarcastic jibe

C) a sincere tribute

D) a mocking taunt

43. Every statement in a notice is attributed to its actual speaker. The notice reports, "Mara claims that the tunnel is open." What does the notice establish by itself?

A) Mara made that claim, but the tunnel's actual condition is not thereby established.

B) The tunnel is definitely open.

C) The tunnel is definitely closed.

D) Mara has personally walked through the tunnel.

44. The sentence "The guide saw a traveler with a telescope" has two possible interpretations. Which information would resolve its grammatical ambiguity?

A) Whether the traveler was standing or sitting

B) Whether the guide had met the traveler before

C) Whether the path continued beyond the traveler

D) Whether the telescope was the guide's viewing instrument or an object accompanying the traveler

45. A critic says a proposal cannot work because its author once made an unrelated spelling mistake. Which additional fact would most directly address whether the proposal can work?

A) The number of people who noticed the spelling mistake

B) The color of the paper on which the proposal appeared

C) Evidence testing the proposal's actual steps and results

D) The critic's preference for shorter proposals

46. The NIMBLE climber moved quickly and lightly across the narrow ledge. NIMBLE most nearly means:

A) motionless

B) careless

C) agile

D) exhausted

47. Only a TENUOUS connection linked the rumor to the actual event. TENUOUS most nearly means:

A) firmly established

B) needlessly complicated

C) widely celebrated

D) weak and uncertain

48. The editor's INCISIVE questions exposed the argument's hidden weakness. INCISIVE most nearly means:

A) politely evasive

B) sharply perceptive

C) carelessly repetitive

D) unusually lengthy

49. The normally DOCILE animal resisted every attempt to lead it away. Which word most nearly expresses the opposite of DOCILE?

A) defiant

B) obedient

C) compliant

D) tractable

50. The water was PRISTINE, untouched by pollutants. Which word most nearly expresses the opposite of PRISTINE in this context?

A) unspoiled

B) pure

C) contaminated

D) untainted

51. Her EBULLIENT greeting overflowed with excitement. Which word most nearly expresses the opposite of EBULLIENT?

A) subdued

B) exuberant

C) animated

D) high-spirited

52. A hinge permits a panel to turn around an axis while restricting other movement. Which device similarly permits one kind of movement while limiting others?

A) A brake slows a wheel by resisting its rotation.

B) A counterweight offsets the weight of a load.

C) A grooved track lets a sliding panel travel along one path while preventing sideways motion.

D) A rough handle improves the user's grip on a tool.

53. A staple joins sheets while leaving each sheet distinct. Which arrangement similarly joins participants without merging their separate identities?

A) A blender mixes several ingredients into a uniform liquid.

B) A furnace melts scraps together into one casting.

C) A shredder cuts several pages into mixed fragments.

D) An alliance lets separate organizations cooperate while each remains an organization.

54. A fingerprint can distinguish one person from others. Which label has the most similar identifying role for instruments manufactured to the same design?

A) A model number shared by every instrument of that design

B) A temperature reading that changes during use

C) A size category containing many different instruments

D) A unique serial number assigned to one individual instrument

55. Which expression belongs with FIRST, MIDDLE, and LAST as a position in an ordered sequence?

A) clockwise

B) next to last

C) upward

D) outward

56. Which statement expresses an obligation rather than a prediction or possibility?

A) The forecast suggests that a report will arrive tomorrow.

B) The rules require every inspector to submit a report.

C) A report may arrive before the office closes.

D) Reports usually arrive during the afternoon.

57. Which word does NOT belong with HEAP, STACK, and CLUSTER as a collection of separate things?

A) pile

B) bunch

C) shard

D) bundle

58. An inspector examines only the top layer of a crate and finds no scratched tiles. She concludes that no tile anywhere in the crate is scratched. Why is that conclusion not established?

A) A top-layer tile cannot ever have a scratch.

B) Finding an undamaged tile proves every lower tile is damaged.

C) The conclusion depends only on the inspector's handwriting.

D) The unexamined lower layers could differ in the condition being generalized.

59. A scan may begin only after an approval is complete. The approval may begin only after a copy is complete. The copy may begin only after the scan is complete. None is complete, and no exceptions or outside records are allowed. What follows?

A) The scan may begin because it was mentioned first.

B) None of the three tasks can begin under these rules.

C) The approval is automatically complete without any work.

D) Any order will work if every task is performed quickly.

60. Call two words linked when they share at least one letter. MINT is linked to TONE, and TONE is linked to SOUR. Which statement is correct?

A) MINT and SOUR are not linked; sharing a letter with an intermediate word does not guarantee a shared letter with each other.

B) MINT and SOUR must be linked because any chain of links guarantees a direct link.

C) All three words are linked simply because they are English words.

D) The links cannot be evaluated without knowing the words' meanings.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. Starting with −2, alternate adding 5 and multiplying by −1. What comes next?
−2, 3, −3, 2, −2, 3, ...

A) −2

B) −3

C) 2

D) 8

62. For n = 1, 2, 3, ... , each entry counts the positive perfect squares not exceeding n. The first nine entries are 1, 1, 1, 2, 2, 2, 2, 2, 3. What is the twentieth entry?

A) 3

B) 16

C) 20

D) 4

63. Each entry is the difference between successive prime numbers, beginning with 2 and 3. What follows?
1, 2, 2, 4, 2, 4, ...

A) 1

B) 2

C) 4

D) 6

64. The nth entry counts EVERY appearance of the digit 9 in the integers from 1 through 10n. For n = 1 through 9, the entries are 1, 2, 3, 4, 5, 6, 7, 8, 10. What is the tenth entry?

A) 10

B) 18

C) 20

D) 19

65. Which integer is even, divisible by 3, and less than 15?

A) 12

B) 9

C) 10

D) 18

66. Positive numbers p and q satisfy pq = 1 and p + q = 5/2. What is 1/p + 1/q?

A) 2/5

B) 5/2

C) 5/4

D) 7/2

67. How many two-digit positive integers equal four times the sum of their digits?

A) 3

B) 5

C) 9

D) 4

68. The positive integers m and n are different.
Quantity A: Their least common multiple
Quantity B: Their greatest common factor

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

69. A first list contains a, b, and c. A second contains a + 7, b − 2, and c − 5.
Quantity A: The first list's arithmetic mean
Quantity B: The second list's arithmetic mean

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

70. The positive integers a and b satisfy a > b. The fractional part of a number is what remains after removing its whole-number part.
Quantity A: The fractional part of a/b
Quantity B: b/a

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

71. A square is inscribed in a circle, with all four vertices on the circle.
Quantity A: The square's area
Quantity B: The circle's area

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

72. In triangle PQR, PQ = PR and QR < PQ.
Quantity A: The angle at Q
Quantity B: The angle at P

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

73. A right prism and an oblique prism have equal base areas and equal perpendicular heights. The oblique prism's side edges lean.
Quantity A: The right prism's volume
Quantity B: The oblique prism's volume

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

74. Start with 7. Subtract 1 to get the second entry, subtract 2 to get the third, and continue increasing the subtraction by 1 each time. Always take the absolute value after subtracting. What comes next?
7, 6, 4, 1, 3, ...

A) 2

B) −2

C) 4

D) 8

75. Each entry is the difference between two consecutive positive cubes, beginning with $2^3 - 1^3$. What comes next?
7, 19, 37, 61, 91, ...

A) 111

B) 125

C) 127

D) 216

76. For each positive integer n, count the different rectangular arrangements of n unit squares with whole-number side lengths. Rotating an arrangement does not make it different. The first six counts are 1, 1, 1, 2, 1, 2. What is the count for n = 36?

A) 4

B) 5

C) 8

D) 10

77. Dividing a positive integer by 5 gives remainder 4. The quotient and remainder add to 7. What is the integer?

A) 15

B) 24

C) 39

D) 19

78. What is the greatest positive integer n for which 3n has exactly two digits and 5n has exactly three digits?

A) 19

B) 33

C) 20

D) 34

79. Use each of the digits 0, 2, and 5 exactly once. Which three-digit number is divisible by 4?

A) 205

B) 250

C) 520

D) 502

80. Two identical calculations are intended to multiply 46 by 18. One mistakenly multiplies 46 by 19; the other mistakenly multiplies 45 by 18. How much greater is the sum of the mistaken results than the sum of the intended results?

A) 28

B) 18

C) 46

D) 64

81. a and b are distinct positive numbers.
Quantity A: a/b + b/a
Quantity B: 2

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

82. a and b are positive integers with a > b.
Quantity A: The greatest common factor of a and b
Quantity B: The greatest common factor of a − b and b

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

83. a, b, and c are integers with a < b < c.
Quantity A: b − a
Quantity B: c − b

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

84. AB is a chord of a circle, and AB does not pass through its center.
Quantity A: The length of AB
Quantity B: The diameter of the circle

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

85. A right rectangular prism has a 3-by-4 base and height 2. A vertical cut through a base diagonal divides it into two triangular prisms.
Quantity A: The sum of the total surface areas of the two pieces
Quantity B: 72 square units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

86. Circle P has a larger radius than circle Q.
Quantity A: The area of one semicircle of P
Quantity B: The area of the entire circle Q

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

87. Each entry after the first is one more than the square of the preceding entry. What comes next?
1, 2, 5, 26, ...

A) 52

B) 676

C) 677

D) 703

88. To obtain each next entry, multiply all the digits in the current entry. What comes next?
77, 49, 36, 18, ...

A) 9

B) 8

C) 18

D) 81

89. Begin with 12345. Reverse the first two digits, then the first three, then the first four, then the first five. Which entry follows?
12345, 21345, 31245, 42135, ...

A) 53142

B) 54213

C) 13524

D) 53124

90. Term n is the greatest number of pieces a flat circular sheet can have after n straight cuts. Arrange every cut to cross all earlier cuts at different interior points. The first four terms are 2, 4, 7, 11. What is term 8?

A) 37

B) 29

C) 36

D) 45

91. The same digit replaces both boxes in 4□ + □7 = 113. What is the digit?

A) 3

B) 6

C) 5

D) 7

92. Define a □ b = a + 2b. How much must be added to (1 □ 2) □ 3 to equal 1 □ (2 □ 3)?

A) 0

B) 3

C) 6

D) 12

93. Start with 1/1. At each step, choose either to add the denominator to the numerator or to add the numerator to the denominator. Which fraction can result after EXACTLY three steps?

A) 2/3

B) 3/2

C) 1/1

D) 5/3

94. a is positive.
Quantity A: $a + \sqrt{(a^2 + 1)}$
Quantity B: 2a

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

95. a < 0 < b, and |a| > b.
Quantity A: a + b
Quantity B: 0

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

96. n is an integer greater than 1. Count 1 and the number itself as positive factors.
Quantity A: The number of positive factors of n
Quantity B: The number of positive factors of n + 1

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

97. In a coordinate plane, a point is reflected first across x = 0 and then across x = 3.
Quantity A: The final x-coordinate minus the original x-coordinate
Quantity B: 6

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

98. Two circular disks have radii 2 and 5 units. Their interiors do not overlap; their boundaries are allowed to touch.
Quantity A: The distance between their centers
Quantity B: 7 units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

99. A triangle has side lengths a, b, and c. The angle opposite c is obtuse.
Quantity A: c^2
Quantity B: $a^2 + b^2$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

100. To obtain each next entry, subtract the greatest power of 2 that does not exceed the current entry. Powers of 2 include 1, 2, 4, 8, 16, 32, and so on. What comes next?
59, 27, 11, 3, ...

A) 0

B) 2

C) 1

D) 4

101. Numbers arrive in this order: 60, 36, 48, 30, 45. After each arrival, record the greatest common factor of ALL numbers received so far. What completes the recorded series?
60, 12, 12, 6, ...

A) 1

B) 5

C) 15

D) 3

102. Each next entry counts how many times the current entry appears among the last FOUR entries, including itself. If fewer than four exist, use all of them. Which TWO entries follow?
1, 1, 2, 1, 3, 1, 2, 1, 2, 2, ...

A) 3, 1

B) 4, 1

C) 2, 3

D) 1, 2

103. A positive whole number ends in 0. Erasing that final zero decreases its value by 243. What was the original number?

A) 27

B) 270

C) 297

D) 2430

104. Two numbers satisfy $a + b = 9$ and $a^2 + b^2 = 45$. What is ab?

A) 36

B) 18

C) 45

D) 81

105. A positive number r satisfies $r + 1 = 1/r$. What is $r^2 + r$?

A) 0

B) 2

C) $r + 1$

D) 1

106. Positive numbers a, b, and c satisfy $ab = 15$, $bc = 21$, and $ac = 35$. What is $a + b + c$?

A) 15

B) 9

C) 21

D) 71

107. x is a nonzero real number.
Quantity A: $|x| + x^2$
Quantity B: x

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

108. n is a positive integer.
Quantity A: The least common multiple of $4n$ and $6n$
Quantity B: $12n$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

109. $0 < x < y < 1$.
Quantity A: xy
Quantity B: $y - x$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

110. A triangle is nondegenerate, so each side is shorter than the sum of the other two.
Quantity A: The longest side
Quantity B: Half the perimeter

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

111. A rectangle and a square have the same positive area. The rectangle's length and width are unequal.
Quantity A: The rectangle's perimeter
Quantity B: The square's perimeter

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

112. Two right circular cylinders have the same radius and different positive heights. Lateral area means only the curved side, excluding the circular ends.
Quantity A: First cylinder's lateral area divided by its volume
Quantity B: Second cylinder's lateral area divided by its volume

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: Nothing Interesting

I hid Dad's birthday present in the bottom drawer of my desk. The problem was that my brother sometimes borrowed things from that desk. To discourage him, I made a label in my most ordinary handwriting: NOTHING INTERESTING.

Then I waited for the drawer to become invisible.

That afternoon, I heard my brother reading the label aloud.

"Why does this one need to tell me?" he asked.

"It contains ordinary things."

He pointed to the two unmarked drawers above it. "What do those contain?"

"Ordinary things too."

He looked pleased, as though I had just supplied the missing piece of a puzzle. I suddenly understood that the label had advertised exactly what I wanted hidden.

After he left, I took it off. The desk looked like a desk again. I was putting the paper in the wastebasket when he returned carrying a small wrapped box.

"Since you've chosen a secret place," he said, "could you keep my present in there too?"

I opened the drawer just wide enough to admit his parcel without revealing mine.

"Don't tell Dad."

"I didn't need you to tell me that."

By dinner, the drawer was private again, although it now had two managers. My brother and I agreed that our improved system required no sign at all.

113. What does the brother bring when he returns to the desk?

A) A new label for the drawer

B) A small wrapped box

C) A borrowed desk lamp

D) A piece of the discarded sign

114. Which statement best describes the story's central development?

A) A father discovers both gifts before his birthday.

B) An attempt to conceal a gift by declaring its hiding place uninteresting instead draws attention and leads to a shared secret.

C) A brother refuses to cooperate with any plan involving presents.

D) Carefully labeling every drawer prevents anyone from noticing a hidden object.

115. Why does the brother point to the two drawers without labels?

A) Their ordinary appearance makes the special label on the bottom drawer seem suspicious.

B) He has forgotten how many drawers the desk contains.

C) He wants to prove that the upper drawers cannot be opened.

D) He already knows the precise contents of every drawer.

116. In "the label had advertised exactly what I wanted hidden," "advertised" most nearly means:

A) offered for sale

B) drawn attention to

C) listed in a printed catalog

D) kept out of sight

117. What makes the label NOTHING INTERESTING ironic?

A) It accurately lists every object in the drawer.

B) It is written by someone who cannot read it.

C) Its effort to discourage interest has the opposite effect.

D) It causes both brothers to forget the birthday.

Informational Passage: Moving Without Legs

An earthworm moves forward without legs by coordinating changes in its body's shape with a grip on the ground. Its body consists of fluid-filled segments surrounded by muscles. Because the fluid does not readily compress, changing one dimension of a segment affects another.

Circular muscles form bands around a segment. When one of these bands contracts, it narrows the segment and makes it longer. Muscles running along the body work differently: their contraction makes segments shorter and thicker. Alternating these actions lets different portions extend and shorten.

Shape changes alone, however, do not guarantee forward travel. Tiny bristles help hold selected parts against the surrounding soil. With a rear portion anchored, a front portion can lengthen forward. The front can then grip while shortening behind it draws more of the body ahead. Coordinated waves of activity pass along the worm rather than moving every segment in exactly the same way at once.

The bristles are not miniature walking legs. They provide contact that allows muscular changes to produce useful movement against the ground. On a very slippery surface, muscles might still change the body's shape while less of that motion carries the entire animal forward. The explanation therefore requires both internal changes and an external grip: one creates reach and pulling movement, while the other prevents those movements from simply sliding back.

118. What helps hold selected portions of the worm against the soil?

A) Tiny bristles

B) Rigid walking legs

C) A hard outer shell

D) Wings along each segment

119. Which statement best summarizes the passage's explanation?

A) Coordinated muscle changes move the worm forward when alternating grips connect those changes to the ground.

B) Every segment becomes longer at the same moment, carrying the worm forward without any grip.

C) Bristles operate as little legs while the muscles remain inactive.

D) The fluid must disappear from each segment before the worm can move.

120. On a surface where the bristles cannot grip effectively, which outcome is best supported?

A) Its muscles must stop contracting entirely.

B) Its fluid-filled segments must immediately become empty.

C) The worm may change shape without moving its whole body forward as effectively.

D) Its bristles must transform into walking legs.

121. When the passage says a circular muscle "contracts," the muscle:

A) signs an agreement

B) becomes a separate segment

C) shortens or tightens

D) stops surrounding the body

122. Why does the author discuss a slippery surface near the end?

A) It replaces the earlier explanation with a claim that worms need wings.

B) It proves that body shape never changes during movement.

C) It clarifies why the earlier muscle actions require effective contact with the ground.

D) It introduces an unrelated history of soil formation.

Humanities Passage: The Missing Line on the Plan

A plan of a public square shows paths meeting at neat right angles. After several months, a narrow track appears diagonally across the grass. No designer drew that line. Repeated footsteps made it visible. Such an informal path records a choice people have made, not merely a route someone intended them to follow.

The worn track can teach a designer something important. Perhaps the paved route takes people far out of their way between a bus stop and an entrance. The diagonal reveals that some users value a shorter connection. Ignoring it would mean overlooking evidence about how the square is actually used.

Should every worn track therefore become a paved path? The question is more difficult than the bare line suggests. A muddy shortcut may serve some walkers while excluding people who need a firm, level surface. Its visible users do not represent everyone who might cross the square. A route may also interfere with another important use of the space.

The track is consequently evidence to investigate, rather than a complete design instruction. A thoughtful response might preserve its useful connection while changing its surface or position. Plans express intentions; worn paths reveal some results. Good design can learn from the difference without pretending that either the plan or the track tells the entire story.

123. How does the diagonal track first become visible?

A) A designer adds it to the original printed plan

B) A new paved road is built before anyone uses it

C) The bus stop is moved to the center of the square

D) Repeated footsteps wear it across the grass

124. Which need does the passage specifically mention when discussing people a muddy shortcut may exclude?

A) A firm, level surface

B) A route with no entrances

C) A square without any other users

D) A printed map in every building

125. Why would counting only the people using the shortcut give an incomplete picture of the square's users?

A) Every person in the square necessarily uses the shortcut.

B) Counting the shortcut's users reveals every person's destination.

C) The original plan guarantees that no one has access needs.

D) People unable or unwilling to use that surface would not appear in that count.

126. As used in "an informal path," "informal" most nearly means:

A) used only during conversations

B) hidden from everyone

C) guaranteed to be unsafe

D) not part of the officially designed layout

127. What is the function of the question "Should every worn track therefore become a paved path?"

A) It asks readers to report an actual construction schedule.

B) It proves the author has no view about path design.

C) It introduces a question the passage never considers again.

D) It challenges an automatic conclusion before the author explains important qualifications.

128. How does the passage develop its argument?

A) It lists identical path designs without comparing them.

B) It contrasts an intended layout with actual use, then explains both the value and limits of the resulting evidence.

C) It gives a construction recipe that ignores users' behavior.

D) It begins by rejecting observation and ends by rejecting all plans.

Literary Passage: More Paper

Grandfather set a postcard on the kitchen table and asked me to write to Emil, whom he had not seen since the ferry service closed. "Just a few words," he said. He began with the weather.

Before I finished the first line, the weather had become the winter when the ferry rope froze. That led to a passenger who carried a live rooster, and the rooster led to Emil's peculiar way of laughing without opening his mouth. Grandfather demonstrated. My pen stopped.

"Should I put that in?" I asked.

"Naturally. He'll remember."

I drew a line through my opening and wrote smaller. Soon my letters leaned against one another like passengers in the old ferry. Grandfather was now describing the day Emil repaired the landing with boards taken from his own shed. This, apparently, required an explanation of the shed.

"We need a smaller story," I said.

Grandfather considered the postcard. "Or more paper."

I found an envelope and several blank sheets. When he began again, I stopped crossing things out. The shed acquired a roof, a smell of apples, and a door that never quite shut.

The postcard remained beside us. It still said only "Dear Emil." For the first time that afternoon, its emptiness did not seem like a problem.

129. Grandfather measures the time since he last saw Emil by referring to which event?

A) The construction of a new shed

B) The purchase of a kitchen table

C) The closing of the ferry service

D) The arrival of the postcard

130. Which statement best captures the significance of the narrator's move from the postcard to longer sheets?

A) The narrator makes room for memories after realizing that the message matters more than fitting the original format.

B) A message becomes clearer whenever all personal details are removed.

C) Grandfather cannot remember enough to finish even a short greeting.

D) The narrator proves that postcards are unsuitable for all correspondence.

131. What does the narrator's decision to stop crossing things out most strongly suggest?

A) The narrator has decided not to send any message.

B) The narrator no longer understands Grandfather's words.

C) The narrator believes Emil would prefer a blank postcard.

D) The narrator now regards Grandfather's connected recollections as worth preserving.

132. What does the comparison of the crowded letters to passengers in the old ferry accomplish?

A) It establishes that the postcard was written aboard a moving ferry.

B) It connects the physical crowding on the postcard with the remembered world Grandfather is describing.

C) It proves that every passenger knew Emil personally.

D) It explains why the ferry service originally closed.

133. How does the narrator's attitude change during the passage?

A) From complete indifference to anger at Emil

B) From admiration for the ferry to fear of traveling

C) From mild impatience with the expanding account to patient interest in its details

D) From eagerness to send a long letter to insistence on a brief postcard

Informational Passage: A Small Dip in Starlight

A distant planet may be too faint to appear separately beside its bright star. Astronomers can sometimes detect it indirectly, however, by recording how much light reaches a telescope from the star over time. If the planet passes between the telescope and the star, it hides a small portion of the star's visible surface. The recorded brightness briefly decreases.

This crossing is called a transit. A planet that returns along the same orbit can produce regularly spaced dips. The spacing supplies evidence about how long one orbit takes. The depth of a dip, under suitable observing conditions, helps estimate the planet's size relative to the star: a larger blocking disk generally hides more of the star's light.

A dip alone does not establish that a planet is present. Another star crossing in front of the observed star can also reduce the received light. Dark regions on a rotating star can produce changing brightness as well. Astronomers therefore examine the pattern and seek other observations that can distinguish these explanations.

The method also has a built-in limitation. An orbit tilted away from the observer's line of sight may never carry its planet across the visible star. Such a planet can exist without producing any transit for that observer. A steady brightness record is therefore not a complete inventory of the star's planets.

134. According to the passage, what produces the decrease in brightness during a planetary transit?

A) The star stops producing light during each orbit.

B) The telescope moves behind the observer.

C) The planet blocks part of the star's visible surface.

D) The planet moves behind the star and adds its own light.

135. What is the primary purpose of this passage?

A) To prove that every dip in brightness identifies a planet

B) To compare all methods of measuring a star's temperature

C) To argue that steady stars cannot have orbiting planets

D) To explain what transit observations can reveal and why their evidence has limits

136. Suppose a planet repeatedly orbits a star, but its orbit never carries it across the star from Earth's viewpoint. What does the passage support?

A) The planet cannot have a regular orbital period.

B) That planet may be missed by a search based only on transits visible from Earth.

C) The star must be too dim to observe.

D) The depth of a dip must measure the planet's mass rather than its size.

137. In the second paragraph, "depth" describes:

A) the distance from the telescope down into the star

B) the amount by which recorded brightness decreases during a dip

C) the number of planets completing one orbit

D) the length of a dark region measured beneath a star's surface

138. Why does the author mention another star and dark regions on a rotating star?

A) They show how all alternatives were eliminated for one particular star.

B) They prove that planets always produce deeper dips than stars.

C) They serve as counterexamples to the assumption that every brightness dip must be caused by a planet.

D) They establish that no star can ever have a steady brightness record.

Informational Passage: Opening a Path Through the Threads

On a simple loom, one set of threads is held lengthwise under tension. These threads are called the warp. A second thread, the weft, travels across them. Weaving requires more than laying the weft on top: the two sets must repeatedly cross over and under one another so that the growing cloth holds together.

For plain weave, the loom separates alternate warp threads into two groups. Lifting one group opens a space between the groups, called a shed. A shuttle carries the weft through that opening. The loom then reverses which group is raised, and the shuttle makes another pass. A warp thread that lay above one weft pass consequently lies below the next. This alternation produces the basic interlacing pattern.

After a pass, the newest weft is pressed toward the cloth already formed. Without this step, successive passes would remain too far apart for the intended fabric. Warp tension helps keep the lengthwise threads in useful positions while the shed opens and closes.

The shuttle transports the crossing thread, but it does not choose the pattern by itself. The order in which groups of warp threads rise and fall determines which crossings occur. More elaborate grouping and lifting arrangements can produce other patterns. In this way, a repeated sequence of openings becomes an arrangement that remains visible after the openings themselves have disappeared.

139. Which term names the threads held lengthwise under tension?

A) Weft

B) Shuttle

C) Shed

D) Warp

140. Why is the newest weft pressed toward the cloth already formed?

A) To keep successive passes from remaining too far apart

B) To keep the shed permanently open for every later pass

C) To make the shuttle choose a different pattern

D) To stop the two groups from ever changing positions

141. Which summary best connects the loom's changing openings with the formation of cloth?

A) The shuttle alone determines every feature of a finished cloth.

B) A loom mainly stretches a completed cloth until its gaps disappear.

C) The crossing pattern depends only on how quickly the shuttle travels.

D) A loom makes interlaced cloth by coordinating thread separation, crosswise passes, and the packing of those passes.

142. If the same warp group stayed raised for every pass, what important feature of plain weave would be missing?

A) The existence of any lengthwise threads

B) Each warp thread's alternation between being above and below successive weft passes

C) The possibility of moving a shuttle across the loom

D) The presence of tension anywhere in the warp

143. In the second paragraph, "shed" means:

A) a temporary opening between two groups of warp threads

B) a small building used to store tools

C) the act of losing a layer of fabric

D) a container attached to the shuttle

144. How does the final paragraph develop the explanation?

A) It distinguishes transporting the weft from controlling the crossing pattern, then connects the repeated motions to the finished result.

B) It rejects the earlier explanation and replaces weaving with an unrelated process.

C) It lists famous weavers in chronological order.

D) It argues that the final cloth cannot reveal anything about the weaving process.

Literary Passage: The Initials in the Cuff

Della found the green glove on the bench outside the skating rink. She had lost one like it that morning, and for a moment the sight seemed to settle everything. She put her cold hand inside.

Two letters scratched her wrist. She turned back the cuff and found J.M. stitched into the wool. She did not remember any initials in her own gloves.

Perhaps those were the maker's initials, she thought. It was a convenient explanation, and she avoided looking at the letters again. She held the glove tighter, as though a firmer grip could make it hers.

At the ticket window, a boy in a gray scarf was asking about a lost glove. "Green," he said. "My grandmother put my initials inside so I wouldn't lose it."

Della could have kept walking. The glove was already on her hand, and nobody had seen her pick it up. The boy was looking into the attendant's tray, not toward the benches.

She pulled the glove off.

"Is this it?"

He checked the cuff. "Yes. Thanks."

There was no speech about honesty. He hurried toward the person waiting by the gate, and Della was left with one cold hand again.

She returned to the bench and looked underneath it. Her own glove was still missing, but now she knew which one she was searching for.

145. What detail first interrupts Della's confidence that the found glove is hers?

A) It is green rather than another color.

B) It has the initials J.M. stitched inside its cuff.

C) It was made without any wool.

D) It has no opening for a hand.

146. Which statement best describes Della's central choice?

A) She trades her own gloves for a warmer pair at the ticket window.

B) She returns a glove she wants to keep after acknowledging evidence that it belongs to someone else.

C) She refuses to look for her missing glove after speaking to the boy.

D) She discovers that every green glove belongs to the same maker.

147. Why does Della initially suggest to herself that J.M. might be the maker's initials?

A) She has already spoken to the boy's grandmother.

B) She wants to identify the maker so she can sell the glove.

C) That possibility lets her treat the unfamiliar initials as compatible with the glove being hers.

D) She remembers ordering a glove labeled J.M. that morning.

148. Calling the maker explanation "convenient" most strongly suggests that it:

A) is located close to the ticket window

B) supports the conclusion Della would prefer to believe

C) makes the glove physically easier to carry

D) has been confirmed by a written receipt

149. What does holding the glove "as though a firmer grip could make it hers" emphasize?

A) A physical rule that changes a glove's owner when it is squeezed

B) The difference between Della's desire to possess the glove and evidence of actual ownership

C) The boy's intention to give Della a present

D) Della's certainty that she had stitched the initials herself

Literary Passage: The Rattle at the Handle

Bracken waited at the far end of the stall while Kira carried in his feed. She set the red bucket beside the door. The horse took a step forward, then lifted his head and stopped.

"Stubborn again," Kira said.

She raised the bucket to move it nearer. Its loose metal handle struck the rim. Bracken stepped backward so quickly that Kira nearly followed him out of surprise.

From the neighboring stall, Tomas looked over the partition. "What happened just before he moved?"

"He decided not to eat."

"Before that."

Kira looked down. When she had picked up the bucket, the handle had fallen against its side. She had heard the sound without attending to it, like hearing her own footsteps.

She waited until Bracken settled. Then, standing where she was, she lifted the handle slightly and let it tap the rim once. His ears turned toward it; his head rose again.

Kira held the handle firmly against the bucket and moved it without a rattle. Bracken watched, stretched his neck, and finally lowered his head to the feed.

"He wasn't refusing the breakfast," she said.

Tomas had already returned to his work. Kira stayed a moment longer. On the feeding list, beside Bracken's name, she crossed out the note she had made yesterday: difficult. In its place she wrote, "Hold the handle still."

150. What immediately happens when the bucket's loose handle strikes its rim?

A) Tomas leaves the stable.

B) Kira spills the feed.

C) Bracken knocks the door down.

D) Bracken steps backward.

151. What is the main significance of Kira's change to the feeding list?

A) She decides that Bracken should never receive breakfast.

B) She proves that all horses dislike red objects.

C) She replaces a judgment about the horse's character with an observation about something she can change.

D) She transfers responsibility for every stable task to Tomas.

152. Why does Kira lift and release the handle again while remaining in the same place?

A) To make Tomas stop working in the neighboring stall

B) To test whether the bucket can hold more feed

C) To persuade Bracken that his breakfast has been removed

D) To see whether the sound, apart from bringing the bucket nearer, is connected with Bracken's reaction

153. What does the comparison between hearing the rattle and hearing her own footsteps emphasize?

A) Kira had registered a familiar sound without considering that it might matter to someone else.

B) The bucket and her shoes made exactly the same noise.

C) Bracken could not hear any sound that Kira could hear.

D) Kira had been deliberately making the loudest possible noise.

154. How is the passage chiefly organized?

A) A list of unrelated feeding rules is followed by a description of a competition.

B) A distant childhood memory interrupts an account of a journey.

C) Several different narrators give contradictory versions without any resolution.

D) An initial judgment is followed by a question, a repeated observation, and a revised explanation.

Informational Passage: Two Sides of a Mountain Range

Two valleys can lie at similar heights on opposite sides of a mountain range yet receive very different amounts of rain. One reason is the route taken by moisture-bearing air. The prevailing wind, meaning the wind that most often comes from a particular direction, carries that air toward one side of the range.

As the air rises along the mountains, it expands and cools. Cooling can cause some of its water vapor to condense into droplets. Clouds may form, and some water can fall as rain or snow before the air crosses the highest ground. The side facing the incoming wind therefore often receives more precipitation.

Beyond the crest, the air descends. Descending air becomes warmer, making condensation less likely. It may also have already lost part of its water on the first side. The valley beyond the range can consequently be much drier, a situation called a rain shadow. A mountain does not need to act like a solid wall that stops every cloud for this contrast to develop.

This explanation describes a recurring pattern, not a promise about every storm. Winds can shift, and other sources of moisture can affect either valley. Nor does every dry region lie in a rain shadow. To decide whether the pattern helps explain a particular place, one must consider both the surrounding landforms and the usual journey of its incoming air.

155. According to the passage, what happens to the air as it rises along a mountain?

A) It expands and cools.

B) It becomes warmer because it is descending.

C) It loses the ability to move horizontally ever again.

D) It must release every bit of water it contains.

156. Which statement best summarizes the passage's explanation of a rain shadow?

A) Mountains stop every cloud, leaving no air able to cross them.

B) Two valleys at equal heights must receive equal rainfall.

C) Air can cool and lose water on the incoming side, then warm as it descends, helping make the opposite side drier.

D) A dry valley always proves that a nearby mountain is its only cause.

157. Suppose two valleys have similar heights but different rainfall. Why would their height alone be insufficient to explain the difference described here?

A) Their positions along the air's route across the mountains can expose them to different stages of the same process.

B) The passage says height never affects any weather process.

C) A prevailing wind must always stop in the first valley.

D) One valley must receive artificial rain supplied by people.

158. As defined in the first paragraph, a "prevailing" wind is one that:

A) never changes its strength

B) always carries the greatest possible amount of water

C) begins only after rain has stopped

D) most often comes from a particular direction

159. What role does the final paragraph play?

A) It limits the earlier generalization and identifies evidence needed to apply it to an actual location.

B) It denies that rising air can ever cool.

C) It turns the explanation into a fictional conversation between two valleys.

D) It lists exact rainfall totals for every mountain range.

Literary Passage: Again

While her blanket turned inside the laundromat dryer, Sylvie hummed the tune her aunt had taught her. She knew where each phrase rested, including the long note near the middle.

A man folding shirts at the next table joined in. At the long note, he slipped in a quick extra note, an ornament that curled around Sylvie's plain tone.

She stopped. "Again?" she said, with the sharp politeness she used when someone gave her the wrong change.

He repeated the phrase.

"There isn't a little note there."

"There is in the way my mother sang it."

Sylvie reached for her phone, then remembered that she had no recording. Her aunt had sung the melody while shelling peas; Sylvie had learned it by joining in.

"Where was your mother from?" she asked.

He named a village on the other side of the hills. Her aunt had grown up near the coast. He sang a little farther, and this time the ending was shorter than Sylvie expected.

"Not in the way I learned it," she said, softening her earlier claim.

The dryer stopped. Neither of them moved toward it.

Sylvie hummed her longer ending. The man tried it, missed a turn, and laughed. She tried his quick extra note. It took a second attempt before her voice found it.

"Again?" she asked. This time she was already leaning closer.

160. How had Sylvie originally learned the tune?

A) By copying a recording on her phone

B) By reading a score at the laundromat

C) By taking lessons from the man folding shirts

D) By joining in while her aunt sang it

161. What does Sylvie's change from "There isn't a little note there" to "Not in the way I learned it" most strongly suggest?

A) She has discovered a recording proving that the man is wrong.

B) She has forgotten the tune her aunt taught her.

C) She now distinguishes her own learned version from a claim about every version of the tune.

D) She believes the extra note is impossible for a human voice.

162. In this passage, the musical "ornament" is:

A) an object hanging from the dryer

B) a written record of the song's owner

C) a brief added note decorating the melody

D) a signal that the tune has ended

163. What is the effect of repeating "Again?" near both the beginning and the end?

A) It proves that Sylvie has learned nothing during the conversation.

B) The same word first challenges a difference and later invites further sharing, revealing Sylvie's changed attitude.

C) It establishes that the dryer repeats the same sound twice.

D) It shows that the man refuses to sing any phrase more than once.

164. Which description best traces the passage's development?

A) A formal lesson ends with a written examination of both singers.

B) A confident correction becomes a comparison of sources and then a mutual attempt to learn unfamiliar details.

C) A disagreement about payment causes both characters to leave.

D) A complete historical account proves where the melody was first composed.

Informational Passage: Following One Grain

Imagine marking a grain of sand near the foot of a moving dune and watching it through many windy afternoons. The dune does not roll forward as one solid block. Its outline changes through the accumulated movement of separate grains.

Wind strong enough to move loose sand can push grains along the ground or lift them into short hops. Some travel up the dune's exposed slope. Near the crest, its highest ridge, they may pass over to the more sheltered side. Sand collecting there can become steep enough to slip downward in small avalanches.

Repeated transfers remove material from the windward side and deposit it beyond the ridge. With a persistent wind direction, the dune's crest and sheltered slope can gradually shift forward. A photograph taken much later may show the familiar shape standing in a new position, although individual grains have followed separate routes and may have stopped at different places.

Our marked grain might become buried partway through this process. Another grain could pass over the ridge sooner. Tracking one grain therefore describes that grain's journey; comparing the dune's outline at different times describes a changing landform. These are related observations, but they answer different questions.

Vegetation or insufficient wind can hold much sand in place. The movement described here depends on grains being loose enough to travel and on wind repeatedly transporting them.

165. What can happen to sand that accumulates on the sheltered side of the dune?

A) It must turn into solid rock immediately.

B) It forces every grain to follow the same route.

C) It prevents any later grain from crossing the ridge.

D) It can slip downward in small avalanches.

166. Which statement best expresses the passage's main idea?

A) A dune can change position through many separate grain movements, so following its shape differs from following one grain.

B) Every grain in a dune moves forward at the same speed.

C) A dune travels only by rolling as a rigid object.

D) A single marked grain always reveals the full history of the entire dune.

167. If the marked grain becomes buried while the dune's ridge continues shifting, what conclusion is best supported?

A) The photograph must show a different kind of material.

B) The dune's changing position does not require every individual grain to keep moving forward with its ridge.

C) No other grain can move until the marked grain reaches the surface.

D) The wind must have stopped permanently.

168. As used here, "crest" means:

A) the buried base of a grain

B) the strongest possible wind

C) the dune's highest ridge

D) a plant anchoring the sand

169. Why does the author introduce an imagined marked grain?

A) To give the reader a particular object to follow before contrasting its route with the dune's changing outline

B) To claim that marked grains behave differently from unmarked grains

C) To explain how paint causes all dunes to move

D) To replace an explanation of dune movement with instructions for photography

Humanities Passage: A Fit That Could Travel

Consider a repairer working with fittings made individually in different workshops. A nut that fits one screw may resist another that looks almost identical. Diameter alone does not settle the matter: the spacing and shape of the threads must match as well. The repairer may spend time altering the replacement until the particular pair works.
An agreed set of dimensions changes the work. A maker can produce a fitting to a stated specification, and a distant repairer can select a replacement using that same specification. Each pair no longer needs its own private history of adjustment. The benefit extends beyond the object itself: people who have never met can coordinate through shared measurements.
Such agreements also demand work. Makers must check their tools, inspect the resulting pieces, and decide how much variation is acceptable. This permitted variation is a tolerance. A dimension written on paper accomplishes little unless production remains within its stated limits. Changing an established system can also leave old stock incompatible with new pieces.
Standardization thus redirects skilled judgment toward selecting dimensions, setting limits, and maintaining repeatable production. A rule can make parts interchangeable within its system while leaving other systems different. Shared dimensions function like a common language for a particular exchange; their usefulness depends on the participants agreeing on what its signs mean.

170. Besides diameter, what must match for the nut and screw described in the opening to fit?

A) The age of the two repairers

B) The color of the specification sheet

C) The location where both objects were first sold

D) The spacing and shape of their threads

171. Which statement best captures what shared specifications make possible and what they require?

A) Writing one dimension guarantees that every fitting in the world will work.

B) Shared specifications can let distant makers and repairers coordinate, provided that production and interpretation follow the agreed limits.

C) Standardization removes every need for skill or inspection.

D) The appearance of a fitting always reveals its complete compatibility.

172. A replacement has the same visible diameter as an old screw, but its specification is unknown. What would the passage caution against assuming?

A) That any measurements could help compare the pieces

B) That individually adjusted fittings ever existed

C) That the replacement occupies physical space

D) That matching appearance alone guarantees a compatible fit

173. In the third paragraph, "tolerance" means:

A) an allowed amount of dimensional variation

B) patience with a repairer's personality

C) the number of workshops in a town

D) a promise that inspection is unnecessary

174. What does comparing shared dimensions to a common language help explain?

A) A fitting must be able to speak before it can be replaced.

B) Only people from the same town can exchange measurements.

C) Both enable coordination when the participants agree on how the shared signs are interpreted.

D) Every language and every fitting system uses exactly the same signs.

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. A transparent panel is divided into four equal rectangles. Exactly half of one rectangle is covered with paper. What fraction of the entire panel is covered?

A) 1/8

B) 1/2

C) 1/4

D) 3/8

176. A gauge reads −12 meters, which is 4 meters lower than the actual water level. The actual level then rises another 3 meters. What is the new actual level?

A) −8 meters

B) −5 meters

C) −13 meters

D) −19 meters

177. One beacon flashes at minutes 0, 8, 16, and so on. Another flashes at minutes 2, 8, 14, and so on. After minute 10, when do they next flash together?

A) Minute 16

B) Minute 24

C) Minute 32

D) Minute 26

178. Which list contains ALL fractions n/12 that are already in lowest terms and satisfy 1/3 < n/12 < 3/4, where n is a positive integer?

A) 5/12 and 7/12

B) 5/12, 6/12, 7/12, and 8/12

C) 4/12, 5/12, and 7/12

D) 7/12 only

179. In the rule y = 3x + 4z, x increases by 4 and z decreases by 3. What happens to y?

A) It stays the same.

B) It increases by 1.

C) It decreases by 1.

D) It increases by 24.

180. Which interval contains exactly the real numbers satisfying |2x − 5| < 3?

A) −4 < x < −1

B) −3 < x < 3

C) x < 1 or x > 4

D) 1 < x < 4

181. A print shop charges an $8 setup fee, $3 each for the first 4 prints, and $2 for each additional print. A bill is $34. How many prints were ordered?

A) 4

B) 7

C) 11

D) 13

182. Two values of x satisfy (x − 1)(x + 4) = 0. Which expression has the same value for both?

A) $x^2 + 3x$

B) x^2

C) $x^2 - 3x$

D) $x + 1$

183. Two identical isosceles right triangles are joined along their hypotenuses, with their interiors on opposite sides of the shared edge. What is the outline of the combined figure?

A) A scalene triangle

B) A pentagon

C) A rhombus with no right angles

D) A square

184. A prism has a total of 18 edges. How many vertices does it have?

A) 6

B) 12

C) 8

D) 18

185. A rectangle measures 5 by 12 units. One diagonal is drawn. What is the perpendicular distance from a corner not on that diagonal to the diagonal?

A) 5 units

B) 60/13 units

C) 12 units

D) 13 units

186. Two cords are measured to the nearest centimeter as 12 centimeters and 18 centimeters. Halfway values are rounded up. Placed end to end without overlap, their actual combined length is L. Which interval is exactly the range of possible L?

A) $29.5 \text{ cm} \le L < 30.5 \text{ cm}$

B) $30 \text{ cm} \le L < 31 \text{ cm}$

C) $29 \text{ cm} \le L < 31 \text{ cm}$

D) $L = 30 \text{ cm}$

187. A 1:50 architectural model has a flat canopy roof with area 24 square centimeters. What is the actual canopy's area in square meters?

A) 0.12 square meter

B) 6 square meters

C) 0.6 square meter

D) 60 square meters

188. A survey records only favorite flavors, with no rankings or numerical scores. Which measure can describe the most typical response?

A) Arithmetic mean

B) Median

C) Numerical range

D) Mode

189. A bag contains one red bead and one blue bead. Draw one at random, return it, and add another bead of the same color. Then draw again. What is the probability that the second color matches the first?

A) 1/4

B) 1/3

C) 2/3

D) 1/2

190. Six bundles have a median mass of 12 kilograms. Which statement MUST be true?

A) At least one bundle weighs exactly 12 kilograms.

B) The arithmetic mean is 12 kilograms.

C) Every bundle weighs 12 kilograms.

D) The third and fourth masses in increasing order total 24 kilograms.

191. Without finding the exact product, which interval must contain 49.8 × 0.201?

A) Between 1 and 3

B) Between 9 and 11

C) Between 19 and 21

D) Between 90 and 110

192. A positive integer leaves remainder 2 when divided by 6. What remainder does its square leave when divided by 6?

A) 0

B) 2

C) 8

D) 4

193. A three-digit palindrome reads the same forward and backward. What is the greatest three-digit palindrome divisible by 18?

A) 828

B) 888

C) 858

D) 898

194. A lift can carry at most 600 kilograms. The cargo already aboard weighs 180 kilograms. Each additional identical crate weighs x kilograms. Which inequality describes adding three crates without exceeding the limit?

A) $180x + 3 \leq 600$

B) $180 + 3x \leq 600$

C) $180 + 3x \geq 600$

D) $3(x + 180) \leq 600$

195. A printer's profit on n posters is P = 4n − (35 + 1.5n) dollars. What is the least whole number of posters that produces a positive profit?

A) 9

B) 15

C) 14

D) 24

196. A student solves x(x − 7) = 0 by dividing both sides by x and reports only x = 7. What is the error?

A) Dividing by x excluded the valid solution x = 0.

B) The student should have divided by 7 instead.

C) The equation has no solutions because its right side is zero.

D) The solution x = 7 must be discarded because it is positive.

197. A rule has the form f(x) = ax + 3, where a is a constant. If f(f(0)) = 9, what is f(4)?

A) 5

B) 9

C) 15

D) 11

198. Point A is at (5, 3). It is rotated 180° about the point (2, −1). What are the coordinates of its image?

A) (−5, −3)

B) (−1, 5)

C) (−1, −5)

D) (1, −5)

199. Two tangents from point P touch a circle at A and B. PA = 3x + 2 and PB = 5x − 8. What is the total length PA + PB?

A) 34

B) 17

C) 10

D) 24

200. ABCDE is a regular pentagon whose vertices are named in order around its boundary. Segments AC and AD are drawn. What is the measure of angle CAD?

A) 18°

B) 54°

C) 36°

D) 72°

201. Runner P takes 5 minutes per kilometer. Runner Q takes 32 seconds per 100 meters. Over the same distance, which runner has the faster pace?

A) Runner Q

B) They have equal paces.

C) Runner P

D) Their paces cannot be compared without their total distances.

202. A metronome clicks 120 times per minute at a steady rate. The first click occurs at time 0. How many seconds elapse before the sixty-fourth click?

A) 30 seconds

B) 31.5 seconds

C) 32 seconds

D) 63 seconds

203. A digital scale adds the same 25-gram error to every reading. A container alone reads 140 grams; the container with flour reads 515 grams. What is the flour's actual mass?

A) 350 grams

B) 400 grams

C) 375 grams

D) 490 grams

204. Which survey question is most likely to encourage agreement through its wording?

A) Which of the two timetables do you prefer?

B) How many times did you use either timetable last week?

C) At what time did your most recent trip begin?

D) Should sensible travelers support the clearly superior new timetable?

205. Three paired observations are (1, 10), (2, 20), and (3, 30). They are changed to (1, 30), (2, 20), and (3, 10). Which feature changes?

A) The mean of the first coordinates

B) The mean of the second coordinates

C) The range of the second coordinates

D) The direction of the association between the two variables

206. A signal passes exactly when relay A works and at least one of relays B and C works. Each relay independently works with probability 1/2. What is the probability that the signal passes?

A) 3/8

B) 1/8

C) 1/2

D) 3/4

207. A dial starts at 0 and advances clockwise by 3/8 of a full turn three times. After passing the starting point once, how far beyond it does the dial finish?

A) 1/8 of a turn

B) 3/8 of a turn

C) 5/8 of a turn

D) 7/8 of a turn

208. How many distinct rational numbers lie strictly between 1/4 and 1/3, when there is no restriction on their denominators?

A) None

B) Infinitely many

C) Exactly one

D) Exactly twelve

209. Positive quantities a and b satisfy a : b = 7 : 5. What percent of a + b is a − b?

A) 28 4/7%

B) 40%

C) 16 2/3%

D) 58 1/3%

210. How many positive integers not exceeding 1,000 are both perfect cubes and numbers with an odd number of positive factors?

A) 3

B) 2

C) 4

D) 10

211. How many real values of x satisfy |x + 6| = −2?

A) Exactly one real value

B) Exactly two real values

C) No real values

D) Every real value

212. Consider E = (x² + 2x)/(x + 2). Which statement correctly simplifies E while preserving its original input restriction?

A) E = x for every real x.

B) E = x for every real x except −2.

C) E = x + 2 for every real x except −2.

D) E = x for every real x except 0.

213. A display has x trays, of which exactly 2 remain empty. Its 36 stones are divided equally among all occupied trays. Which expression gives the stones in each occupied tray, where x > 2?

A) 36/x − 2

B) (36 − 2)/x

C) 36/(x − 2)

D) 36/(x + 2)

214. For every real x, a rule satisfies f(x + 3) = f(x) − 4. If f(2) = 9, what is f(−7)?

A) −3

B) −19

C) 13

D) 21

215. ABCD is a parallelogram with vertices in that order. A = (−1, 2), B = (3, 2), and D = (1, 5). What are the coordinates of C?

A) (1, 2)

B) (3, 5)

C) (−3, −1)

D) (5, 5)

216. A circular sector has central angle 150° and arc length 5π centimeters. What is the circle's radius?

A) 3 centimeters

B) 12 centimeters

C) 6 centimeters

D) 15 centimeters

217. Two unit squares have vertices (0, 0), (1, 0), (1, 1), (0, 1) and (1, 1), (2, 1), (2, 2), (1, 2). A tight rubber band surrounds both squares without dipping inward between them. What is its perimeter?

A) 4 + 2√2 units

B) 8 units

C) 6 units

D) 4 + √2 units

218. Machine A starts at time 0 and needs 7 minutes 30 seconds to make one part. Machine B starts 2 minutes 10 seconds later and needs 4 minutes 50 seconds for another part. Assembly needs both finished parts and takes 40 seconds. What is the earliest completion time measured from time 0?

A) 7 minutes 30 seconds

B) 8 minutes 10 seconds

C) 7 minutes 40 seconds

D) 15 minutes 10 seconds

219. A fabric's width shrinks by 8% in its first wash. After washing, 2 centimeters at each side are folded out of the visible width. What prewash width is needed for a visible width of 180 centimeters?

A) 1.80 meters

B) 2.00 meters

C) 1.84 meters

D) 1.96 meters

220. The data are 2, 2, 4, and 7. Which value can be added so that the range stays the same but the median changes?

A) 2

B) 1

C) 3

D) 8

221. Five measurements have both mean 10 and median 10. One measurement equal to 10 is removed. Which statement must be true?

A) Both the mean and median must stay 10.

B) Only the median must stay 10.

C) Neither the mean nor the median is guaranteed to stay 10.

D) The mean stays 10, but the median may change.

222. Six equally spaced points on a circle are labeled A through F in order. Three different points are selected, with every three-point set equally likely. What is the probability that they form an equilateral triangle?

A) 1/5

B) 1/3

C) 1/2

D) 1/10

223. In the decimal 0.406, the tenths digit is increased by 1 and the thousandths digit is decreased by 6. What is the net increase in the number's value?

A) 0.106

B) 0.040

C) 0.094

D) 0.960

224. What is the least positive integer n for which n^3 is divisible by 200?

A) 10

B) 5

C) 20

D) 200

225. A 250-gram mixture contains 40% copper by mass. One fifth of the mixture is removed, with the removed portion having the same composition. Then 50 grams of pure copper is added. What percent of the final mixture is copper?

A) 52%

B) 40%

C) 60%

D) 72%

226. Squaring $\sqrt{(x + 2)} = x$ gives $x + 2 = x^2$, whose candidate solutions are 2 and −1. Which statement gives the solution of the ORIGINAL equation?

A) $x = 2$ only

B) $x = -1$ only

C) $x = 2$ or $x = -1$

D) There is no real solution.

227. A row contains n solar panels. One support stands between each adjacent pair, and one additional support stands at each end. The combined number of panels and supports is 55. How many panels are there?

A) 26

B) 28

C) 54

D) 27

228. Which choice gives ALL real solutions of $|2x - 1| = |x + 2|$?

A) $x = 3$ only

B) $x = 3$ and $x = -1/3$

C) $x = -3$ and $x = 1/3$

D) $x = -1$ and $x = 3$

229. A device's F command moves f centimeters right and its B command moves b centimeters left, with f and b fixed and positive. Starting from 0 each time, FFB ends at 17 and FBB ends at 1. Where does FFFB end?

A) 18 centimeters to the right

B) 22 centimeters to the right

C) 28 centimeters to the right

D) 34 centimeters to the right

230. A plane slices off one corner of a cube by crossing the three edges that meet at that corner. What shape is the newly exposed flat face?

A) A triangle

B) A square

C) A rectangle

D) A hexagon

231. A circular ring has inner radius 4 and outer radius 6. A quarter of the ring is cut out along two perpendicular radii. What is the full perimeter of that quarter-ring piece?

A) 5π + 20 units

B) 3π + 4 units

C) 10π + 4 units

D) 5π + 4 units

232. Portable screens form three sides of a rectangular work area; a straight wall forms the fourth. Exactly 24 meters of screens are available. What is the greatest area that can be enclosed?

A) 36 square meters

B) 72 square meters

C) 64 square meters

D) 144 square meters

233. A carton of material lists its coverage as 24 square meters. What quantity does that measurement describe?

A) The distance around the carton

B) The surface area the material can cover

C) The material's thickness

D) The space enclosed inside the carton

234. A uniform elastic cord is 1.2 meters long and has mass 30 grams. It is stretched uniformly to 1.5 meters without changing mass. What is the mass of a 30-centimeter segment of the stretched cord?

A) 7.5 grams

B) 20 grams

C) 24 grams

D) 6 grams

235. On an original plan, 1 centimeter represents 5 meters in both directions. A copy stretches only the horizontal direction to 150% of its original size. A rectangular room measures 3 centimeters horizontally and 4 centimeters vertically on the copy. What is its actual area?

A) 300 square meters

B) 200 square meters

C) 450 square meters

D) 40 square meters

236. A delivery log records damaged-item counts as 0, 2, blank, and 1. The blank means no count was recorded. Which interpretation is justified?

A) The blank proves that the count was zero.

B) The recorded zero must be removed before any summary is made.

C) The blank is missing information and cannot automatically be treated as zero.

D) The blank must equal the largest recorded count.

237. Five measurements have minimum 0, median 4, and maximum 8. Which feature is NOT determined by that summary?

A) The range

B) Whether 4 occurs among the measurements

C) Whether any measurement is greater than 8

D) The arithmetic mean

238. Three switches are independently set up or down, each with probability 1/2. A light is on when a majority are up. One designated switch is then reversed. What is the probability that the light's on-or-off status changes?

A) 1/4

B) 3/8

C) 1/2

D) 3/4

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which sentence correctly sets off a person being directly addressed within a question?

A) Will, you Captain check the mooring before we leave?

B) Will you Captain, check the mooring before we leave?

C) Will you, Captain check the mooring before we leave?

D) Will you, Captain, check the mooring before we leave?

240. Several geese cross the mud. Which phrase correctly identifies the footprints as belonging to those geese collectively?

A) the geese's footprints

B) the geeses' footprints

C) the goose's footprints

D) the geese footprint's

241. Which version correctly uses US quotation marks and commas for a quotation inside a quotation?

A) "I heard "all clear," but the gate remained closed," Ana reported.

B) "I heard 'all clear', but the gate remained closed," Ana reported.

C) "I heard 'all clear,' but the gate remained closed," Ana reported.

D) "I heard 'all clear,' but the gate remained closed", Ana reported.

242. Which sentence correctly distinguishes a title before a name from the same word naming a general occupation?

A) professor Doran is a Professor of engineering.

B) Professor Doran is a Professor of Engineering.

C) professor Doran is a professor of Engineering.

D) Professor Doran is a professor of engineering.

243. World History II is the official title of a course. Which sentence correctly uses that title and then mentions the general subject?

A) Her schedule lists world history II, and she enjoys studying World History.

B) Her schedule lists World History II, and she enjoys studying world history.

C) Her schedule lists World History II, and she enjoys studying World History.

D) Her schedule lists World history II, and she enjoys studying world History.

244. Choose the correctly spelled name of the long-necked animal.

A) giraffe

B) girafe

C) girraffe

D) giraff

245. Which spelling correctly completes the sentence? "The beetle's ___ shell seemed to change color as it turned."

A) irridescent

B) iridescant

C) iridesent

D) iridescent

246. Which sentence places only so that it identifies the front door as the sole locked door?

A) The front door was only locked.

B) The front door was locked only at night.

C) Only the front door was locked.

D) The only front door was locked.

247. Choose the pair that correctly completes the sentence: "The supervisor made the contractor ___ the railing and persuaded the painter ___ it."

A) to inspect; finish

B) inspecting; to finish

C) inspect; to finish

D) inspect; finishing

248. The engineer has not inspected the cables. Which reply correctly says that the apprentice also has not inspected them?

A) So has the apprentice.

B) Neither has the apprentice.

C) Neither have the apprentice.

D) Neither does the apprentice.

249. Choose the word meaning that something serves two purposes: "The foldout panel has a ___ role."

A) duel

B) duly

C) dull

D) dual

250. Choose the pair that means "reduce the appeal" in the first blank and "divert attention" in the second: "The scratch may ___ from the frame's beauty; the loud noise may ___ the viewer."

A) distract; detract

B) detract; distract

C) detract; detract

D) distract; distract

Editing Passage: Putting a Kite Away

(1) [Topic sentence goes here.] (2) Packing a damp kite immediately can leave its fabric trapped in moisture until the next outing. (3) Spread it out to dry before folding or closing its storage bag. (4) Before removing the rods, note how each length fits into its sleeve or connector. (5) Then remove them and group the matching pieces for later assembly. (6) Some kite festivals include performances after sunset. (7) Fold the dry fabric loosely, keeping sharp rod ends from pressing directly against it. (8) Place small connectors in a separate pouch so they cannot disappear among the folds. (9) The packed parts should be protected and easy to recognize when the kite is assembled again.

251. Which sentence best supplies the missing topic sentence?

A) Putting a kite away properly protects its material and keeps its parts ready for the next use.

B) A kite festival should always begin after sunset.

C) The best way to store a kite is to hide every connector inside a tight fold.

D) Matching rods are unnecessary once a kite has flown.

252. Which sentence should be deleted because it does not develop the storage procedure?

A) Sentence 3

B) Sentence 7

C) Sentence 6

D) Sentence 8

253. Which revision best combines sentences 4 and 5 while preserving the necessary order of actions?

A) Note how the rods fit before removing them, then group matching pieces for later assembly.

B) Remove and mix all rods before deciding whether their original positions matter.

C) Because rods fit every sleeve equally well, remove them without noting their positions.

D) Group the rods permanently and discard their connectors before noting how the kite fits together.

254. Which version correctly punctuates the spelled-out number in the sentence?

A) The invoice lists seventy-four bolts.

B) The invoice lists seventy four bolts.

C) The invoice lists seventy, four bolts.

D) The invoice lists seventy: four bolts.

255. Which sentence correctly uses commas when an introductory yes is followed by a negative response?

A) Yes I checked the valve, but no I did not open it.

B) Yes, I checked the valve, but no, I did not open it.

C) Yes, I checked the valve but, no I did not open it.

D) Yes I checked, the valve but no, I did not open it.

256. Which sentence correctly distinguishes the chemical symbol Co, meaning cobalt, from the formula CO, meaning carbon monoxide?

A) The container marked CO holds cobalt; the detector marked Co detects carbon monoxide.

B) The container marked co holds cobalt; the detector marked co detects carbon monoxide.

C) The container marked cO holds cobalt; the detector marked cO detects carbon monoxide.

D) The container marked Co holds cobalt; the detector marked CO detects carbon monoxide.

257. Which sentence correctly distinguishes the agency abbreviation NASA from radar, an acronym that has become an ordinary technology word?

A) Nasa engineers tested the Radar.

B) NASA engineers tested the radar.

C) NASA Engineers tested the RADAR.

D) nasa engineers tested the radar.

258. Choose the correctly spelled name of the fruit with many seeds enclosed in a thick rind.

A) pomegranate

B) pomegranite

C) pomagranate

D) pomegranet

259. Which spelling correctly completes the sentence? "The underground passage twisted like a ___."

A) labarynth

B) labirynth

C) labyrinth

D) labyrenth

260. Choose the correctly spelled word for a substance that removes moisture from its surroundings.

A) dessicant

B) desiccant

C) desicant

D) desiccint

261. Which sentence correctly uses a singular count noun after every?

A) Every spools have a numbered tag.

B) Every spool have a numbered tag.

C) Every spool has a numbered tag.

D) Every spools has a numbered tag.

262. Choose the pair that correctly completes the sentence: "I remember ___ the hatch yesterday, but please remember ___ it again tonight."

A) to lock; locking

B) lock; locked

C) locked; to locking

D) locking; to lock

263. "The sail needs mending" means that the sail requires repair. Which revision preserves that meaning in a standard passive infinitive construction?

A) The sail needs to mend.

B) The sail needs being mend.

C) The sail needs to be mending.

D) The sail needs to be mended.

264. A STATUTE is a written law, while STATURE concerns height or standing. Which sentence uses the appropriate word?

A) The new statute establishes the maximum permitted load.

B) The new stature establishes the maximum permitted load.

C) The climber's statute made the low doorway difficult to use.

D) The cabinetmaker measured his apprentice's statute in centimeters.

265. Which sentence correctly uses all together or altogether to mean that the travelers waited in one group?

A) Altogether, the travelers waited for an hour.

B) The travelers' delay was altogether unexpected.

C) The travelers gave up waiting altogether.

D) The travelers waited all together under the awning.

Editing Passage: The Price of Flexibility

(1) A travel price should be considered together with the conditions attached to it. (2) On the fictional Cedar Line, a fixed ticket costs less but cannot be refunded after purchase. (3) A flexible ticket costs more and can be refunded before departure. (4) [Transition], the flexible ticket is not automatically the better choice for every traveler. (5) Someone whose plans are settled may gain little from paying for a refund option. (6) A traveler awaiting confirmation of an appointment faces a different possibility. (7) Paying more initially could protect that person from losing the entire fare if the journey becomes unnecessary. (8) [Concluding sentence goes here.]

266. Which transition best fills the blank in sentence 4?

A) For example

B) As a result of its lower price

C) Nevertheless

D) In exactly the same way

267. Where should this sentence be added? "That appointment might be canceled before the planned departure."

A) Immediately before sentence 1

B) Immediately after sentence 6

C) Immediately after sentence 2

D) Immediately after sentence 5

268. Which sentence best concludes the passage without adding an unsupported rule?

A) The better purchase depends on how likely the traveler is to need the protection that the higher price buys.

B) Every traveler should buy the cheapest ticket regardless of its restrictions.

C) The flexible ticket is worthwhile only when the trip is certain to take place.

D) A higher initial fare guarantees that a traveler will take the journey.

269. Which version correctly treats the action after the quoted speech as a separate sentence rather than as a speech tag?

A) "I'll take it," Nessa lifted the parcel.

B) "I'll take it." Nessa lifted the parcel.

C) "I'll take it", Nessa lifted the parcel.

D) "I'll take it." Nessa, lifted the parcel.

270. Which sentence correctly punctuates size, shape, and material adjectives that build on one another rather than operate as coordinate descriptions?

A) The carpenter restored a small, round, oak table.

B) The carpenter restored a small round oak table.

C) The carpenter restored a small round, oak table.

D) The carpenter restored a small, round oak table.

271. Which version places semicolons between three complete options while keeping each option's explanatory relative clause intact?

A) The choices were to rent a van; which offered more room, to repair the cart; which needed new wheels, or to postpone the delivery; which required everyone's consent.

B) The choices were; to rent a van, which offered more room; to repair the cart, which needed new wheels or to postpone the delivery, which required everyone's consent.

C) The choices were to rent a van, which offered; more room, to repair the cart, which needed; new wheels, or to postpone the delivery, which required; everyone's consent.

D) The choices were to rent a van, which offered more room; to repair the cart, which needed new wheels; or to postpone the delivery, which required everyone's consent.

272. Which pair gives the two lowercase words that need capitals in this sentence? "During october, we stayed in a canadian village near the border."

A) October; Canadian

B) Village; Border

C) October; Border

D) Canadian; Village

273. The parenthesized items below continue ONE sentence rather than start separate sentences. Which version uses appropriate capitalization?

A) The work involves (a) cleaning the joints, (b) replacing the seals, and (c) checking the pressure.

B) The work involves (a) Cleaning the joints, (b) replacing the seals, and (c) checking the pressure.

C) The work involves (a) cleaning the joints, (b) Replacing the seals, and (c) checking the pressure.

D) The work involves (a) Cleaning the joints, (b) Replacing the seals, and (c) Checking the pressure.

274. Which spelling correctly names the activity of traveling over snow on skis?

A) skiiing

B) sking

C) skiing

D) skiying

275. Choose the correctly spelled word for a figure used to display clothing.

A) manequin

B) mannekin

C) manniquin

D) mannequin

276. Choose the word that correctly completes the sentence: "The report contains ___ reliable information."

A) a

B) some

C) an

D) several

277. Rosa began copying a chart at 10:00 yesterday and continued without stopping until 10:10. Which sentence correctly describes her activity at 10:05 yesterday?

A) At 10:05, Rosa will be copying the chart.

B) At 10:05, Rosa has copying the chart.

C) At 10:05, Rosa was copying the chart.

D) At 10:05, Rosa were copying the chart.

278. The keeper said to Ari, "Do not disturb the seals." Which sentence correctly reports that negative command?

A) The keeper told Ari not to disturb the seals.

B) The keeper told Ari do not disturb the seals.

C) The keeper told Ari to not disturbed the seals.

D) The keeper told Ari not disturbing the seals.

279. Choose the pair that correctly completes the sentence: "The editor gave a clear ___ for what the team considered a ___ decision."

A) rational; rationale

B) rationale; rationale

C) rationale; rational

D) rational; rational

280. Choose the words, in order, for the sense of taste, a board for mixing paints, and a platform for stacked freight.

A) palette; palate; pallet

B) pallet; palette; palate

C) palate; palette; pallet

D) palate; pallet; palette

281. Choose the pair meaning an organized plan of practice in the first blank and a military unit in the second: "The drummer maintained a strict ___ before joining the ___."

A) regiment; regimen

B) regimen; regimen

C) regiment; regiment

D) regimen; regiment

Editing Passage: Keeping the Condition

(1) A quotation can copy words accurately while changing the reach of the speaker's claim. (2) A headline containing only "The method saves time" leaves out that requirement. (3) Suppose a technician says, "The method saves time if the parts have already been sorted." (4) This shortened wording could make an unprepared reader expect an immediate benefit before doing the sorting work. (5) In the original statement, sorting is the condition under which time is saved. (6) Keeping that condition, or explaining it fairly beside the quotation, protects the statement from being expanded beyond what its speaker actually claimed.

282. Which sentence order most clearly presents the claim, explains its condition, shows the misleading shortened version, and concludes with a remedy?

A) 1, 2, 4, 5, 3, 6

B) 1, 3, 5, 2, 4, 6

C) 1, 5, 3, 2, 6, 4

D) 3, 5, 6, 4, 2, 1

283. Which title best represents the passage's purpose?

A) Preserving the Limits of a Quoted Claim

B) The Fastest Way to Sort Every Part

C) Why All Short Quotations Are False

D) Replacing a Technician's Words with a New Opinion

284. Which sentence correctly encloses the explanatory noun phrase describing the courier?

A) The courier a cyclist carrying a blue satchel, waited outside.

B) The courier, a cyclist carrying a blue satchel waited outside.

C) The courier, a cyclist carrying a blue satchel, waited outside.

D) The courier a cyclist, carrying a blue satchel waited outside.

285. The original notice says, "The north gate opens at dawn." Which report integrates an exact quoted phrase and correctly places the final period in American English?

A) The notice says the north gate opens "at dawn."

B) The notice says the north gate opens, "at dawn."

C) The notice says the north gate opens "at dawn".

D) The notice says the north gate opens "after dawn."

286. The semicolon below joins two independent clauses in ONE sentence. Which version uses correct capitalization?

A) The switch was off; The screen stayed dark.

B) The Switch was off; the Screen stayed dark.

C) The switch was off; the screen stayed dark.

D) the switch was off; the screen stayed dark.

287. Central processing unit is an ordinary descriptive term, not a formal proper name. Which sentence correctly writes that term and its initial-letter abbreviation?

A) The Central Processing Unit (CPU) coordinates the calculations.

B) The central processing unit (cpu) coordinates the calculations.

C) The central processing unit (CPU) coordinates the calculations.

D) The Central processing unit (Cpu) coordinates the calculations.

288. Choose the correctly spelled name of the melon.

A) cantalope

B) cantaloupe

C) canteloupe

D) cantaloupp

289. Which spelling correctly completes the sentence? "The orchestra built to a powerful ___."

A) cresscendo

B) cresendo

C) creschendo

D) crescendo

290. Choose the correctly spelled name of the flowering plant.

A) chrysanthemum

B) crysanthemum

C) chrysanthimum

D) chrysanthemun

291. Which completion makes this a full sentence? "What the diver discovered beneath the ledge ___."

A) after the dive

B) surprised the crew

C) near the broken anchor

D) before the boat returned

292. Which sentence uses must to express a confident inference about a COMPLETED PAST action?

A) They must take the early boat to arrive on time.

B) They must be taking the boat right now.

C) They must have taken the early boat; their berth is empty.

D) They must take care of their boat.

293. Choose the pair that correctly completes the sentence in formal written English: "I would rather ___ here tonight, but I'd rather my guest ___ early tomorrow."

A) stay; arrived

B) to stay; arrived

C) stay; arriving

D) staying; to arrive

294. Choose the pair that correctly completes the sentence: "The map compares long-term ___ conditions; the novel reaches its ___ scene."

A) climactic; climatic

B) climatic; climatic

C) climactic; climactic

D) climatic; climactic

295. Choose the pair that correctly completes the sentence: "In the legend, the oracle will ___; the villagers later recall the ___."

A) prophecy; prophesy

B) prophesy; prophecy

C) prophesy; prophesy

D) prophecy; prophecy

296. Choose the words meaning the RATE of such events in the first blank and individual EVENTS in the second: "The report measured the ___ of faults and described several separate ___."

A) incidence; incidents

B) incidents; incidence

C) incidence; incidence

D) incidents; incidents

Editing Passage: A Guide That Survives Copying

(1) A transit guide uses three colors to distinguish its routes. (2) In a particular black-and-white photocopy, those colors become nearly the same gray. (3) Giving each route a different line pattern as well as a color would

preserve another visible distinction. (4) One route could use a solid line, another a dashed line, and the third a dotted line. (5) The key should show the same patterns beside the corresponding route names. (6) This will make it easier to follow. (7) The aim is to keep the route information usable when color can no longer do the separating.

297. Which added sentence would provide the most direct evidence for the recommendation in sentence 3?

A) Readers said the colored cover looked attractive.

B) The print shop finished thirty copies before lunch.

C) The original colors were selected from popular travel posters.

D) In a test of the grayscale copy, readers distinguished the solid and dashed routes even where their gray tones matched.

298. Which revision of sentence 6 most clearly identifies the proposed change and its benefit?

A) It will improve the thing that people use.

B) This proves that all color printing should be abandoned.

C) Readers will follow it because it is easier than the other ones.

D) Distinct line patterns will help readers follow each route in a copy that has no usable color.

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. B	2. A	3. A	4. D	5. C	6. A	7. D	8. C
9. A	10. C	11. B	12. C	13. D	14. B	15. B	16. C
17. A	18. D	19. A	20. B	21. D	22. C	23. B	24. A
25. B	26. C	27. C	28. D	29. B	30. D	31. C	32. C
33. B	34. D	35. A	36. A	37. D	38. A	39. B	40. D
41. B	42. C	43. A	44. D	45. C	46. C	47. D	48. B
49. A	50. C	51. A	52. C	53. D	54. D	55. B	56. B
57. C	58. D	59. B	60. A				

Quantitative Skills

61. B	62. D	63. B	64. C	65. A	66. B	67. D	68. A
69. C	70. D	71. B	72. A	73. C	74. A	75. C	76. B
77. D	78. B	79. C	80. A	81. A	82. C	83. D	84. B
85. C	86. D	87. C	88. B	89. D	90. A	91. B	92. C
93. D	94. A	95. B	96. D	97. C	98. D	99. A	100. C
101. D	102. A	103. B	104. B	105. D	106. A	107. A	108. C
109. D	110. B	111. A	112. C				

Reading Comprehension

113. B	114. B	115. A	116. B	117. C	118. A	119. A	120. C
121. C	122. C	123. D	124. A	125. D	126. D	127. D	128. B
129. C	130. A	131. D	132. B	133. C	134. C	135. D	136. B
137. B	138. C	139. D	140. A	141. D	142. B	143. A	144. A
145. B	146. B	147. C	148. B	149. B	150. D	151. C	152. D
153. A	154. D	155. A	156. C	157. A	158. D	159. A	160. D
161. C	162. C	163. B	164. B	165. D	166. A	167. B	168. C
169. A	170. D	171. B	172. D	173. A	174. C		

Mathematics

175. A	176. B	177. C	178. A	179. A	180. D	181. C	182. A
183. D	184. B	185. B	186. C	187. B	188. D	189. C	190. D
191. B	192. D	193. A	194. B	195. B	196. A	197. D	198. C
199. A	200. C	201. C	202. B	203. C	204. D	205. D	206. A

207. A	208. B	209. C	210. A	211. C	212. B	213. C	214. D
215. D	216. C	217. A	218. B	219. B	220. A	221. D	222. D
223. C	224. A	225. A	226. A	227. D	228. B	229. C	230. A
231. D	232. B	233. B	234. D	235. B	236. C	237. D	238. C

Language

239. D	240. A	241. C	242. D	243. B	244. A	245. D	246. C
247. C	248. B	249. D	250. B	251. A	252. C	253. A	254. A
255. B	256. D	257. B	258. A	259. C	260. B	261. C	262. D
263. D	264. A	265. D	266. C	267. B	268. A	269. B	270. B
271. D	272. A	273. A	274. C	275. D	276. B	277. C	278. A
279. C	280. C	281. D	282. B	283. A	284. C	285. A	286. C
287. C	288. B	289. D	290. A	291. B	292. C	293. A	294. D
295. B	296. A	297. D	298. D				

Detailed Explanations

Verbal Skills

1. HSPT-V16-001 - Answer B: extremely hungry
Ravenous means extremely hungry; the missed meal and long walk support that meaning rather than a mood or a need for sleep.

2. HSPT-V16-002 - Answer A: in a state of serious disrepair
Dilapidated describes a structure that has deteriorated through age or neglect, as the damaged roof and boards demonstrate.

3. HSPT-V16-003 - Answer A: intensely passionate
Fervent describes strong, passionate feeling; it characterizes the intensity of the appeal rather than its volume or clarity.

4. HSPT-V16-004 - Answer D: brisk
Sluggish movements are slow and lacking energy; brisk movements are quick and energetic, reversing the intended description.

5. HSPT-V16-005 - Answer C: optional
Obligatory means required by a rule or duty; optional means left to a person's choice rather than required.

6. HSPT-V16-006 - Answer A: cheerful
Morose describes a gloomy or sullen mood; cheerful expresses the contrasting positive, pleasant state of mind.

7. HSPT-V16-007 - Answer D: changes the language while preserving the stated meaning
Both change a form of expression while preserving what is expressed: the measured quantity in one case and the meaning in the other.

8. HSPT-V16-008 - Answer C: An email filter separates marked junk messages while admitting other messages.
Both apply a criterion to separate admitted material from excluded material; indiscriminate deletion, copying, and rearrangement do not perform that selective function.

9. HSPT-V16-009 - Answer A: A slow-motion replay expands time intervals while preserving the movement's path and order.
Both expand one dimension to reveal detail while preserving other relationships; reversal, cropping, and invention instead alter or remove those relationships.

10. HSPT-V16-010 - Answer C: An intermission
An intermission extends between a beginning and an end; the other choices identify particular moments rather than durations.

11. HSPT-V16-011 - Answer B: melting; freezing; evaporating
Melting, freezing, and evaporating change physical state; the other groups include actions that merely move an object or substance.

12. HSPT-V16-012 - Answer C: lender; borrower
A loan necessarily connects a lender and borrower; writers, singers, and travelers can act without publishers, conductors, or guides.

13. HSPT-V16-013 - Answer D: The agreement comes from a shared source rather than separate checks.
Copies can repeat one source's information or error; their agreement does not show that separate sources independently checked the report.

14. HSPT-V16-014 - Answer B: At least one repaired instrument passed inspection but was not delivered.
A false universal claim requires a counterexample meeting its conditions: repaired, passed inspection, and not delivered.

15. HSPT-V16-015 - Answer B: The three rules cannot all be satisfied.
The last two rules make P, Q, and R agree, while the first requires P and Q to disagree, creating a contradiction.

16. HSPT-V16-016 - Answer C: unnecessary
Superfluous means more than is needed; the second identical diagram adds nothing necessary to the already clear instructions.

17. HSPT-V16-017 - Answer A: impossible to erase
Indelible describes a mark that cannot be removed or erased; the failed attempts to wash away the stain establish that sense.

18. HSPT-V16-018 - Answer D: filled with deep hostility
Rancorous describes bitterness and lasting resentment, which the neighbors continue to feel after their argument has ended.

19. HSPT-V16-019 - Answer A: dense
Sparse vegetation is thinly distributed; dense vegetation grows thickly, reversing the description of how closely the plants are grouped.

20. HSPT-V16-020 - Answer B: revocable
Irrevocable means unable to be withdrawn or canceled; revocable means that withdrawal or cancellation remains possible.

21. HSPT-V16-021 - Answer D: exuberant
Staid suggests restraint and seriousness; exuberant suggests lively, overflowing energy, providing the strongest contrast in atmosphere.

22. HSPT-V16-022 - Answer C: Changing every yes on a fixed checklist to no and every no to yes
Both exchange opposed values at each position while preserving the arrangement; rearranging, deleting, and copying perform different operations.

23. HSPT-V16-023 - Answer B: A saved checkpoint that lets a game resume from recorded progress
Both preserve a stopping position for later continuation; the alternatives describe content, report time, or rank items without preserving progress.

24. HSPT-V16-024 - Answer A: Using a temporary spacer, then removing it so a permanent piece can fit in the reserved opening
Both use a temporary occupant to establish space for a later occupant; decoration, measurement, and polishing do not create space by removal.

25. HSPT-V16-025 - Answer B: receipt
A receipt acknowledges a completed payment; an estimate, quotation, or bid states a proposed price before the transaction is completed.

26. HSPT-V16-026 - Answer C: accidental; inadvertent; unplanned
Accidental, inadvertent, and unplanned all exclude deliberate planning; each other group includes at least one word that explicitly indicates intention.

27. HSPT-V16-027 - Answer C: kernel
A kernel is an inner seed or edible core; shell, skin, and hull name removable outer coverings, like the three given words.

28. HSPT-V16-028 - Answer D: Actions taken from opposite sides can move the same door in the same direction.
Pushing from one side can produce the same door movement as pulling from the other; the instructions apply to different positions.

29. HSPT-V16-029 - Answer B: The sealing rule was violated, regardless of the lock's result.
Passing both checks is necessary for a seal; a failed lining defeats that requirement whether the lock passed or failed.

30. HSPT-V16-030 - Answer D: The indicator is lit.
If the engine is not idling, Ravi's statement requires a lit indicator; if it is idling, Lena's statement requires a lit indicator.

31. HSPT-V16-031 - Answer C: deeply thoughtful
Pensive describes quiet, serious thought; Lena's silent consideration supports that meaning rather than pride, movement, or impatience.

32. HSPT-V16-032 - Answer C: charmingly old-fashioned
Quaint suggests an appealingly unusual or old-fashioned quality; the traditional windows and shutters establish that intended sense.

33. HSPT-V16-033 - Answer B: harshly biting
Caustic language is sharply critical or cutting; the effect on the performer supports the figurative meaning rather than a chemical one.

34. HSPT-V16-034 - Answer D: effective
Futile describes an effort that cannot produce a useful result; effective describes one that accomplishes its intended purpose.

35. HSPT-V16-035 - Answer A: public
Clandestine means deliberately kept secret; public describes something openly known, while the other choices preserve the secrecy.

36. HSPT-V16-036 - Answer A: imaginative
In this context, prosaic means lacking imagination or distinction; imaginative reverses that judgment, whereas the remaining choices reinforce it.

37. HSPT-V16-037 - Answer D: The height of a note on a musical staff indicates its pitch.
Both assign meaning to a mark's position; timing a beat, signaling an entrance, and protecting an instrument serve other functions.

38. HSPT-V16-038 - Answer A: A potter examines each fired test tile and adjusts the next firing according to the result.
Feedback uses an observed result to adjust a later action; fixed schedules, unused records, and a customer's selection do not form that loop.

39. HSPT-V16-039 - Answer B: A temporary pass stops permitting entry when its stated period ends unless it is extended.
Both grant a benefit for a limited period and require further authorization to continue it; records and physical marks do not expire that way.

40. HSPT-V16-040 - Answer D: lair
A lair is an animal's den or hiding place; crates, canisters, and cartons are manufactured containers for storing or transporting objects.

41. HSPT-V16-041 - Answer B: untie; disconnect; unwrap
Untie, disconnect, and unwrap undo tying, connecting, and wrapping; the other groups include actions that need not reverse an earlier operation.

42. HSPT-V16-042 - Answer C: a sincere tribute
A sincere tribute expresses genuine respect; sneering, sarcastic, and mocking expressions belittle or ridicule their targets rather than honor them.

43. HSPT-V16-043 - Answer A: Mara made that claim, but the tunnel's actual condition is not thereby established.
Accurately attributing a claim establishes who said it, not whether its content is true or how the speaker obtained it.

44. HSPT-V16-044 - Answer D: Whether the telescope was the guide's viewing instrument or an object accompanying the traveler
With a telescope can describe how the guide saw or what accompanied the traveler; the other details do not identify that phrase's role.

45. HSPT-V16-045 - Answer C: Evidence testing the proposal's actual steps and results
Testing the proposal addresses its operation directly; an unrelated error and the other personal or presentation details do not establish whether its method succeeds.

46. HSPT-V16-046 - Answer C: agile
Nimble describes quick, light, coordinated movement; agile preserves that meaning without suggesting recklessness, stillness, or fatigue.

47. HSPT-V16-047 - Answer D: weak and uncertain
A tenuous connection is weak or slight and provides little support; the phrase concerns the strength of the link rather than its popularity.

48. HSPT-V16-048 - Answer B: sharply perceptive
Incisive questions penetrate to an important issue with clear, sharp understanding; exposing the hidden weakness demonstrates that perceptiveness.

49. HSPT-V16-049 - Answer A: defiant
Docile means readily guided or controlled; defiant describes resistance to control, while the other choices retain the original cooperative meaning.

50. HSPT-V16-050 - Answer C: contaminated
Pristine here means clean and unspoiled; contaminated indicates the presence of unwanted substances and directly reverses that description.

51. HSPT-V16-051 - Answer A: subdued
Ebullient means visibly overflowing with enthusiasm; subdued describes a restrained, quiet manner rather than an energetic display of excitement.

52. HSPT-V16-052 - Answer C: A grooved track lets a sliding panel travel along one path while preventing sideways motion.
Both guide movement by allowing a particular path and restricting departures from it; braking, balancing, and gripping perform different functions.

53. HSPT-V16-053 - Answer D: An alliance lets separate organizations cooperate while each remains an organization.
The staple and alliance establish connections among parts that retain their identities; blending, melting together, and shredding erase or disrupt those distinctions.

54. HSPT-V16-054 - Answer D: A unique serial number assigned to one individual instrument
A unique serial number distinguishes an individual unit; model and size labels group units, while a temperature reading describes a changing condition.

55. HSPT-V16-055 - Answer B: next to last
Next to last locates an item within an ordered sequence; clockwise, upward, and outward describe directions of movement rather than sequence positions.

56. HSPT-V16-056 - Answer B: The rules require every inspector to submit a report.
Require states what inspectors are obligated to do; the other statements describe expected, possible, or customary events without imposing a duty.

57. HSPT-V16-057 - Answer C: shard
A shard is one broken piece of a solid object; piles, bunches, and bundles are collections, like heaps, stacks, and clusters.

58. HSPT-V16-058 - Answer D: The unexamined lower layers could differ in the condition being generalized.
The observations establish the condition of the inspected layer, but they supply no guarantee about unexamined tiles elsewhere in the crate.

59. HSPT-V16-059 - Answer B: None of the three tasks can begin under these rules.
Every possible first task requires another task already to be complete; with no completed starting record, the circular prerequisites block every first step.

60. HSPT-V16-060 - Answer A: MINT and SOUR are not linked; sharing a letter with an intermediate word does not guarantee a shared letter with each other.
MINT shares letters with TONE, and TONE shares O with SOUR, but MINT and SOUR share none; the defined relation is not transitive.

Quantitative Skills

61. HSPT-Q16-001 - Answer B: −3
Adding 5 produced the final 3, so the next operation is multiplication by −1, giving −3.

62. HSPT-Q16-002 - Answer D: 4
The positive squares not exceeding 20 are 1, 4, 9, and 16; therefore the entry for n = 20 is 4.

63. HSPT-Q16-003 - Answer B: 2
The last listed gap is 17 − 13 = 4; the next prime is 19, so the following gap is 19 − 17 = 2.

64. HSPT-Q16-004 - Answer C: 20
Through 100, a 9 occurs ten times in the units place and ten times in the tens place; 99 contributes to both counts.

65. HSPT-Q16-005 - Answer A: 12
Twelve meets all three conditions; 9 is odd, 10 is not divisible by 3, and 18 exceeds the stated bound.

66. HSPT-Q16-006 - Answer B: 5/2
Combining the reciprocals gives (p + q)/(pq); substituting the given sum and product gives (5/2)/1 = 5/2.

67. HSPT-Q16-007 - Answer D: 4
Writing the digits as a and b gives 10a + b = 4(a + b), so b = 2a; a can be 1, 2, 3, or 4.

68. HSPT-Q16-008 - Answer A: Quantity A is greater.
The common factor is at most the smaller integer, while the common multiple is at least the larger; the integers are different, so the inequality is strict.

69. HSPT-Q16-009 - Answer C: The two quantities are equal.
The changes sum to 7 − 2 − 5 = 0, so both lists have the same total and the same number of entries.

70. HSPT-Q16-010 - Answer D: The relationship cannot be determined from the information given.
For a = 3, b = 2, Quantity A is 1/2 < 2/3; for a = 5, b = 3, it is 2/3 > 3/5.

71. HSPT-Q16-011 - Answer B: Quantity B is greater.
The circle contains the whole square and additional regions outside its sides, so its area is strictly greater.

72. HSPT-Q16-012 - Answer A: Quantity A is greater.
The angle at Q lies opposite PR, which is longer than QR opposite the angle at P; a longer side has a larger opposite angle.

73. HSPT-Q16-013 - Answer C: The two quantities are equal.
Each prism's volume is base area times perpendicular height; leaning side edges do not change either given quantity.

74. HSPT-Q16-014 - Answer A: 2
The last step used |1 − 4| = 3. The next step subtracts 5, giving |3 − 5| = 2.

75. HSPT-Q16-015 - Answer C: 127
The last given entry is $6^3 - 5^3 = 91$; the next is $7^3 - 6^3 = 343 - 216 = 127$.

76. HSPT-Q16-016 - Answer B: 5
The unordered side pairs are 1 × 36, 2 × 18, 3 × 12, 4 × 9, and 6 × 6, giving five arrangements.

77. HSPT-Q16-017 - Answer D: 19
The quotient is 7 − 4 = 3; reconstructing the dividend gives 5 × 3 + 4 = 19.

78. HSPT-Q16-018 - Answer B: 33
The digit conditions give 10 ≤ 3n ≤ 99 and 100 ≤ 5n ≤ 999, so 20 ≤ n ≤ 33.

79. HSPT-Q16-019 - Answer C: 520
Divisibility by 4 depends on the final two digits; only 520 ends in a two-digit multiple of 4, namely 20.

80. HSPT-Q16-020 - Answer A: 28
The first error adds 46, while the second subtracts 18; together the errors change the intended sum by 46 − 18 = 28.

81. HSPT-Q16-021 - Answer A: Quantity A is greater.
Subtracting 2 gives $(a - b)^2/(ab)$. The numerator and denominator are positive because a and b are positive and unequal.

82. HSPT-Q16-022 - Answer C: The two quantities are equal.
A number divides both a and b exactly when it divides both a − b and b, so the greatest common factors agree.

83. HSPT-Q16-023 - Answer D: The relationship cannot be determined from the information given.
For 1, 2, 5, the first gap is smaller; for 1, 4, 5, it is larger. Ordering alone does not determine gap sizes.

84. HSPT-Q16-024 - Answer B: Quantity B is greater.
A diameter is the longest chord; because AB does not pass through the center, it is strictly shorter than the diameter.

85. HSPT-Q16-025 - Answer C: The two quantities are equal.
The original surface area is 52. The cut has area 5 × 2 = 10 and creates two exposed faces, making the sum 52 + 20 = 72.

86. HSPT-Q16-026 - Answer D: The relationship cannot be determined from the information given.
If Q has radius 2, P with radius 2.2 gives a smaller semicircle; P with radius 4 gives a larger one.

87. HSPT-Q16-027 - Answer C: 677
Apply the stated square-then-add rule to 26: $26^2 + 1 = 676 + 1 = 677$.

88. HSPT-Q16-028 - Answer B: 8
The successive digit products are 7 × 7 = 49, 4 × 9 = 36, 3 × 6 = 18, and 1 × 8 = 8.

89. HSPT-Q16-029 - Answer D: 53124
The next operation reverses all five digits of 42135, placing them in the order 5, 3, 1, 2, 4.

90. HSPT-Q16-030 - Answer A: 37
Cut k can add k pieces because its k − 1 crossings divide it into k segments. Eight cuts therefore give 1 + 1 + 2 + ... + 8 = 37.

91. HSPT-Q16-031 - Answer B: 6
The units sum must end in 3, so the missing digit plus 7 is 13. The digit is 6, and 46 + 67 = 113.

92. HSPT-Q16-032 - Answer C: 6
The left expression is 5 □ 3 = 11; the right is 1 □ 8 = 17. Their difference is 17 − 11 = 6.

93. HSPT-Q16-033 - Answer D: 5/3
Changing the numerator, then denominator, then numerator gives 1/1 → 2/1 → 2/3 → 5/3. The other choices do not occur after three steps.

94. HSPT-Q16-034 - Answer A: Quantity A is greater.
Because $a^2 + 1 > a^2$ and a is positive, $\sqrt{(a^2 + 1)} > a$. Adding a makes Quantity A greater than 2a.

95. HSPT-Q16-035 - Answer B: Quantity B is greater.
The negative number has greater magnitude than the positive number, so their sum is negative and therefore less than zero.

96. HSPT-Q16-036 - Answer D: The relationship cannot be determined from the information given.
For n = 3, the factor counts are 2 and 3; for n = 4, they are 3 and 2. Either quantity can be greater.

97. HSPT-Q16-037 - Answer C: The two quantities are equal.
An original x-coordinate t becomes −t, then 6 + t after reflection across x = 3; the net displacement is always 6 units right.

98. HSPT-Q16-038 - Answer D: The relationship cannot be determined from the information given.
The centers are 7 units apart when the disks touch externally, but can be farther apart when separated. Equality and a greater distance are both possible.

99. HSPT-Q16-039 - Answer A: Quantity A is greater.
An obtuse angle makes the opposite side longer than the hypotenuse of a right triangle with the same two adjacent sides, so $c^2 > a^2 + b^2$.

100. HSPT-Q16-040 - Answer C: 1
The greatest power of 2 not exceeding 3 is 2, so the next entry is 3 − 2 = 1.

101. HSPT-Q16-041 - Answer D: 3
After the fourth arrival the common factor is 6; including 45 changes it to the greatest common factor of 6 and 45, which is 3.

102. HSPT-Q16-042 - Answer A: 3, 1
The window 2, 1, 2, 2 contains three 2s, giving 3. The updated window 1, 2, 2, 3 contains one 3, giving 1.

103. HSPT-Q16-043 - Answer B: 270
Let the shortened number be n; the original is 10n. Their difference is 9n = 243, so n = 27 and the original is 270.

104. HSPT-Q16-044 - Answer B: 18
Squaring a + b gives $81 = a^2 + 2ab + b^2 = 45 + 2ab$. Thus 2ab = 36 and ab = 18.

105. HSPT-Q16-045 - Answer D: 1
Multiplying both sides of r + 1 = 1/r by the nonzero number r gives $r^2 + r = 1$ directly.

106. HSPT-Q16-046 - Answer A: 15
The ratio (ab)(bc)/(ac) gives $b^2 = 9$, so b = 3. Then a = 5 and c = 7, making their sum 15.

107. HSPT-Q16-047 - Answer A: Quantity A is greater.
If x is positive, Quantity A adds a positive square to x. If x is negative, Quantity A is positive while Quantity B is negative.

108. HSPT-Q16-048 - Answer C: The two quantities are equal.
A multiple 4kn is divisible by 6n when 4k is divisible by 6; the least positive such k is 3, giving 12n.

109. HSPT-Q16-049 - Answer D: The relationship cannot be determined from the information given.
With x = 1/4 and y = 1/2, the product is smaller; with x = 3/5 and y = 4/5, the product is larger.

110. HSPT-Q16-050 - Answer B: Quantity B is greater.
If c is the longest side, $c < a + b$. Adding c gives $2c < a + b + c$, so c is less than half the perimeter.

111. HSPT-Q16-051 - Answer A: Quantity A is greater.
For rectangle sides a and b, the square side is $\sqrt{(ab)}$. The perimeter difference is $2(a + b - 2\sqrt{(ab)}) = 2(\sqrt{a} - \sqrt{b})^2$, which is positive.

112. HSPT-Q16-052 - Answer C: The two quantities are equal.
Lateral area is $2\pi rh$ and volume is $\pi r^2 h$; dividing gives $2/r$. Their common radius therefore makes both ratios equal despite different heights.

Reading Comprehension

113. HSPT-R16-001 - Answer B: A small wrapped box
The brother returns carrying a small wrapped box and asks the narrator to keep his present in the same hiding place.

114. HSPT-R16-002 - Answer B: An attempt to conceal a gift by declaring its hiding place uninteresting instead draws attention and leads to a shared secret.
The label attracts the brother's curiosity, and the failed concealment becomes a cooperative arrangement for hiding both gifts.

115. HSPT-R16-003 - Answer A: Their ordinary appearance makes the special label on the bottom drawer seem suspicious.
The contrast exposes the oddity of announcing that only one drawer is uninteresting, giving him a reason to suspect something special inside it.

116. HSPT-R16-004 - Answer B: drawn attention to
The label attracts curiosity about the hidden drawer; advertised is figurative here and does not involve selling anything.

117. HSPT-R16-005 - Answer C: Its effort to discourage interest has the opposite effect.
The narrator expects the label to reduce attention, but its unusual presence instead prompts the brother to investigate.

118. HSPT-R16-006 - Answer A: Tiny bristles
The passage identifies tiny bristles as structures that help selected parts grip the surrounding soil during coordinated movement.

119. HSPT-R16-007 - Answer A: Coordinated muscle changes move the worm forward when alternating grips connect those changes to the ground.
The passage explains forward travel through alternating extension, shortening, and anchoring, emphasizing that muscular changes and ground contact work together.

120. HSPT-R16-008 - Answer C: The worm may change shape without moving its whole body forward as effectively.
The final paragraph states that shape changes can continue on a slippery surface while producing less forward movement because grip is reduced.

121. HSPT-R16-009 - Answer C: shortens or tightens
Contracts describes the muscle tightening around the segment, narrowing it even though that narrowing makes the segment itself longer.

122. HSPT-R16-010 - Answer C: It clarifies why the earlier muscle actions require effective contact with the ground.
The slippery case removes reliable grip while allowing shape changes, revealing why grip is a necessary partner in the movement already described.

123. HSPT-R16-011 - Answer D: Repeated footsteps wear it across the grass
The opening says repeated footsteps create the diagonal track even though the designer did not draw that route.

124. HSPT-R16-012 - Answer A: A firm, level surface
The third paragraph contrasts users who can take a muddy shortcut with people who need a firm, level surface.

125. HSPT-R16-013 - Answer D: People unable or unwilling to use that surface would not appear in that count.
The passage notes that a muddy track can exclude some users, so observations of its current users omit those excluded people.

126. HSPT-R16-014 - Answer D: not part of the officially designed layout
The path develops through repeated use rather than being drawn in the original design, which establishes this contextual meaning of informal.

127. HSPT-R16-015 - Answer D: It challenges an automatic conclusion before the author explains important qualifications.
The rhetorical question shifts from recognizing useful evidence to examining why that evidence alone cannot determine the final design.

128. HSPT-R16-016 - Answer B: It contrasts an intended layout with actual use, then explains both the value and limits of the resulting evidence.
The opening contrasts plan and track; the remaining paragraphs explain what the track reveals and why additional needs still matter.

129. HSPT-R16-017 - Answer C: The closing of the ferry service
The opening sentence directly says Grandfather had not seen Emil since the ferry service closed; the other events do not establish that interval.

130. HSPT-R16-018 - Answer A: The narrator makes room for memories after realizing that the message matters more than fitting the original format.
The move from cramped postcard writing to uncrossed letter pages shows acceptance of Grandfather's expansive memories rather than insistence on the original limit.

131. HSPT-R16-019 - Answer D: The narrator now regards Grandfather's connected recollections as worth preserving.
The narrator keeps the added details once more paper is available, indicating a changed appreciation of the memories rather than withdrawal from the task.

132. HSPT-R16-020 - Answer B: It connects the physical crowding on the postcard with the remembered world Grandfather is describing.
The simile makes cramped handwriting vivid while drawing its image from the ferry memories; it supplies no literal account of the service's closure.

133. HSPT-R16-021 - Answer C: From mild impatience with the expanding account to patient interest in its details
Crossing out and requesting a smaller story show initial impatience; keeping the shed's details and accepting the empty postcard show later interest.

134. HSPT-R16-022 - Answer C: The planet blocks part of the star's visible surface.
The first paragraph explains that the intervening planet hides a small portion of the star's visible surface, reducing the light received.

135. HSPT-R16-023 - Answer D: To explain what transit observations can reveal and why their evidence has limits
The passage explains the signal, the information it can supply, competing explanations, and the viewing limitation, covering both usefulness and limits.

136. HSPT-R16-024 - Answer B: That planet may be missed by a search based only on transits visible from Earth.
The last paragraph states that an unfavorable orbital tilt can prevent visible transits even when a planet exists, limiting that search method.

137. HSPT-R16-025 - Answer B: the amount by which recorded brightness decreases during a dip
Depth concerns how far the brightness reading falls; the surrounding sentences connect a larger blocking disk with a greater reduction in received light.

138. HSPT-R16-026 - Answer C: They serve as counterexamples to the assumption that every brightness dip must be caused by a planet.
These examples supply alternative causes of reduced light, supporting the author's caution about treating one observed effect as proof of a planet.

139. HSPT-R16-027 - Answer D: Warp
The first paragraph explicitly names the lengthwise threads the warp; the weft crosses them, the shuttle carries it, and the shed is an opening.

140. HSPT-R16-028 - Answer A: To keep successive passes from remaining too far apart
The third paragraph states that pressing brings successive passes together; without it, they remain too widely separated for the intended fabric.

141. HSPT-R16-029 - Answer D: A loom makes interlaced cloth by coordinating thread separation, crosswise passes, and the packing of those passes.
The passage connects the shed's changing groups, the shuttle's passes, and pressing the weft as coordinated parts of cloth formation.

142. HSPT-R16-030 - Answer B: Each warp thread's alternation between being above and below successive weft passes
The second paragraph makes the over-under alternation depend on reversing the raised group; without reversal, that alternating relationship would not occur.

143. HSPT-R16-031 - Answer A: a temporary opening between two groups of warp threads
The passage defines shed as the space opened between separated warp groups, rather than its familiar meaning of a small building.

144. HSPT-R16-032 - Answer A: It distinguishes transporting the weft from controlling the crossing pattern, then connects the repeated motions to the finished result.
The paragraph separates the shuttle's carrying role from the lifting sequence's pattern-making role, then explains how temporary motions leave a lasting arrangement.

145. HSPT-R16-033 - Answer B: It has the initials J.M. stitched inside its cuff.
Finding J.M. inside the cuff conflicts with Della's memory of unmarked gloves and leads her to seek another explanation; the color itself matches.

146. HSPT-R16-034 - Answer B: She returns a glove she wants to keep after acknowledging evidence that it belongs to someone else.
Della first explains away the unfamiliar initials, but the boy's matching description leads her to return the glove despite still needing her own.

147. HSPT-R16-035 - Answer C: That possibility lets her treat the unfamiliar initials as compatible with the glove being hers.
Della wants the glove to be hers, and the maker explanation neutralizes a detail that otherwise conflicts with that wish; no memory supports it.

148. HSPT-R16-036 - Answer B: supports the conclusion Della would prefer to believe
Convenient describes how well the explanation serves Della's wish, while her avoidance of the initials shows that its truth has not been established.

149. HSPT-R16-037 - Answer B: The difference between Della's desire to possess the glove and evidence of actual ownership
The imagined power of her grip exposes wishful thinking: holding an object firmly cannot establish that it belongs to her.

150. HSPT-R16-038 - Answer D: Bracken steps backward.
The narration directly connects the handle striking the rim with Bracken stepping backward; the other events are not described.

151. HSPT-R16-039 - Answer C: She replaces a judgment about the horse's character with an observation about something she can change.
Difficult labels the horse, while Hold the handle still records a practical response to the observed rattle, reflecting Kira's revised understanding.

152. HSPT-R16-040 - Answer D: To see whether the sound, apart from bringing the bucket nearer, is connected with Bracken's reaction
Keeping her position while reproducing the rattle separates the sound from the earlier approach, helping Kira identify which change draws the horse's attention.

153. HSPT-R16-041 - Answer A: Kira had registered a familiar sound without considering that it might matter to someone else.
Own footsteps illustrate a sound noticed automatically but easily ignored, explaining why Kira initially overlooked the rattle as a possible cause.

154. HSPT-R16-042 - Answer D: An initial judgment is followed by a question, a repeated observation, and a revised explanation.
Kira first calls Bracken stubborn, Tomas redirects her attention, and her test of the handle leads to a more specific explanation and instruction.

155. HSPT-R16-043 - Answer A: It expands and cools.
The second paragraph explicitly says rising air expands and cools; it does not require complete water loss or permanent loss of horizontal motion.

156. HSPT-R16-044 - Answer C: Air can cool and lose water on the incoming side, then warm as it descends, helping make the opposite side drier.
The passage links rising and cooling, possible precipitation, and descending and warming to explain a common rainfall contrast while qualifying its universality.

157. HSPT-R16-045 - Answer A: Their positions along the air's route across the mountains can expose them to different stages of the same process.
The opening contrast is explained through the air's rising and descending path, showing that equal valley heights do not imply equal moisture conditions.

158. HSPT-R16-046 - Answer D: most often comes from a particular direction
The first paragraph directly defines prevailing by the direction from which a wind most often comes, not by constant strength or maximum moisture.

159. HSPT-R16-047 - Answer A: It limits the earlier generalization and identifies evidence needed to apply it to an actual location.
Shifting winds and other moisture sources qualify the pattern; the final sentence then calls for considering landforms and the air's usual route.

160. HSPT-R16-048 - Answer D: By joining in while her aunt sang it
The passage explicitly says Sylvie learned by joining her aunt's singing while her aunt shelled peas; it also states that she has no recording.

161. HSPT-R16-049 - Answer C: She now distinguishes her own learned version from a claim about every version of the tune.
The revised wording limits her claim to her own learning after hearing the man's different family source; it no longer declares his version universally wrong.

162. HSPT-R16-050 - Answer C: a brief added note decorating the melody
The surrounding sentence identifies the ornament as a quick extra note curling around the longer tone, establishing its musical rather than physical decorative sense.

163. HSPT-R16-051 - Answer B: The same word first challenges a difference and later invites further sharing, revealing Sylvie's changed attitude.
The first Again carries sharp correction; the last accompanies leaning closer after exchanging versions, turning the repeated word into an invitation.

164. HSPT-R16-052 - Answer B: A confident correction becomes a comparison of sources and then a mutual attempt to learn unfamiliar details.
Sylvie moves from correcting the note to discussing where each version came from, and both characters finally try elements of the other's singing.

165. HSPT-R16-053 - Answer D: It can slip downward in small avalanches.
The second paragraph states that accumulating sand can become steep enough to slip in small avalanches; the other proposed results are not stated.

166. HSPT-R16-054 - Answer A: A dune can change position through many separate grain movements, so following its shape differs from following one grain.
The passage combines grain transport with the distinction between an individual grain's route and the larger landform's changing position.

167. HSPT-R16-055 - Answer B: The dune's changing position does not require every individual grain to keep moving forward with its ridge.
The text allows one grain to stop or become buried while transfers elsewhere reshape the dune, separating individual motion from movement of the outline.

168. HSPT-R16-056 - Answer C: the dune's highest ridge
The passage directly identifies the crest as the highest ridge and then describes grains passing over it to the sheltered side.

169. HSPT-R16-057 - Answer A: To give the reader a particular object to follow before contrasting its route with the dune's changing outline
Following one grain provides a concrete viewpoint; the later comparison shows why that individual route cannot stand for every change in the landform.

170. HSPT-R16-058 - Answer D: The spacing and shape of their threads
The opening explicitly says that thread spacing and shape must match as well as diameter; the other details are not given as fitting requirements.

171. HSPT-R16-059 - Answer B: Shared specifications can let distant makers and repairers coordinate, provided that production and interpretation follow the agreed limits.
The passage develops both the coordination enabled by common dimensions and the inspection, tolerances, and agreement needed to make that coordination reliable.

172. HSPT-R16-060 - Answer D: That matching appearance alone guarantees a compatible fit
The opening explains that similar-looking pieces can differ in thread spacing and shape, so diameter or appearance alone cannot establish compatibility.

173. HSPT-R16-061 - Answer A: an allowed amount of dimensional variation
The author defines tolerance as permitted variation and connects it to keeping actual production within the limits of a stated dimension.

174. HSPT-R16-062 - Answer C: Both enable coordination when the participants agree on how the shared signs are interpreted.
The analogy emphasizes shared interpretation among participants, showing why an agreed system helps distant people coordinate without implying universal compatibility.

Mathematics

175. HSPT-M16-001 - Answer A: 1/8
One rectangle is 1/4 of the panel; covering half of that rectangle covers 1/2 × 1/4 = 1/8 of the whole.

176. HSPT-M16-002 - Answer B: −5 meters
Correcting a reading that is 4 meters too low gives −8 meters; the later 3-meter rise gives −5 meters.

177. HSPT-M16-003 - Answer C: Minute 32
They coincide at minute 8 and then every least common multiple of 8 and 6, or 24 minutes; the next coincidence is minute 32.

178. HSPT-M16-004 - Answer A: 5/12 and 7/12
The strict bounds give n = 5, 6, 7, or 8; only 5 and 7 share no common factor with 12.

179. HSPT-M16-005 - Answer A: It stays the same.
The change in y is 3 × 4 + 4 × (−3) = 12 − 12 = 0, so y stays unchanged.

180. HSPT-M16-006 - Answer D: 1 < x < 4
The absolute-value inequality means −3 < 2x − 5 < 3; adding 5 and dividing by 2 gives 1 < x < 4.

181. HSPT-M16-007 - Answer C: 11
Setup and the first four prints cost $20; the remaining $14 buys seven additional prints, giving 4 + 7 = 11.

182. HSPT-M16-008 - Answer A: $x^2 + 3x$
The solutions are 1 and −4; $x^2 + 3x$ equals 4 for each, while the other expressions give different values at those inputs.

183. HSPT-M16-009 - Answer D: A square
The four outer legs are equal, and the joined 45-degree angles form right angles at the shared edge's endpoints, producing a square.

184. HSPT-M16-010 - Answer B: 12
For an n-sided base, a prism has 3n edges and 2n vertices; 3n = 18 gives n = 6 and 12 vertices.

185. HSPT-M16-011 - Answer B: 60/13 units
The diagonal is 13 units and the triangle's area is 30; using the diagonal as base gives 13h/2 = 30, so h = 60/13.

186. HSPT-M16-012 - Answer C: 29 cm ≤ L < 31 cm
The actual lengths lie in [11.5, 12.5) and [17.5, 18.5); adding the bounds gives 29 cm ≤ L < 31 cm.

187. HSPT-M16-013 - Answer B: 6 square meters
Areas scale by 50² = 2,500, giving 60,000 square centimeters; dividing by 10,000 gives 6 square meters.

188. HSPT-M16-014 - Answer D: Mode
The mode identifies the most frequent category; mean, median, and numerical range require numerical values or an ordering that the flavors do not have.

189. HSPT-M16-015 - Answer C: 2/3
After either possible first draw, the bag contains two beads matching that color and one other bead, so the matching probability is 2/3.

190. HSPT-M16-016 - Answer D: The third and fourth masses in increasing order total 24 kilograms.
For six values, the median is the average of the third and fourth; an average of 12 requires their sum to be 24.

191. HSPT-M16-017 - Answer B: Between 9 and 11
The factors are close to 50 and 0.2, whose product is 10; more precisely, 49 × 0.2 < 49.8 × 0.201 < 50 × 0.21.

192. HSPT-M16-018 - Answer D: 4
Write the integer as 6k + 2. Its square is $36k^2 + 24k + 4$, whose first two terms are divisible by 6.

193. HSPT-M16-019 - Answer A: 828
Divisibility by 18 requires an even final digit and a digit sum divisible by 9. With outer digits 8, only middle digit 2 works.

194. HSPT-M16-020 - Answer B: 180 + 3x ≤ 600
The existing 180 kilograms is counted once, three crates contribute 3x, and at most permits equality but excludes totals above 600.

195. HSPT-M16-021 - Answer B: 15
Profit simplifies to 2.5n − 35. Positive profit requires n > 14, so the least whole number of posters is 15.

196. HSPT-M16-022 - Answer A: Dividing by x excluded the valid solution x = 0.
The original product is zero when x = 0 or x = 7; division by x assumes x is nonzero and loses the first solution.

197. HSPT-M16-023 - Answer D: 11
First f(0) = 3, so f(f(0)) = f(3) = 3a + 3 = 9. Thus a = 2 and f(4) = 11.

198. HSPT-M16-024 - Answer C: (−1, −5)
The center must be the midpoint of A and its image: the image is (2 × 2 − 5, 2 × (−1) − 3) = (−1, −5).

199. HSPT-M16-025 - Answer A: 34
Tangents from the same exterior point have equal lengths: 3x + 2 = 5x − 8 gives x = 5, so both lengths are 17.

200. HSPT-M16-026 - Answer C: 36°
Each pentagon angle is 108°. Isosceles triangles ABC and AED each have a 36° base angle at A, leaving 108° − 36° − 36° = 36°.

201. HSPT-M16-027 - Answer C: Runner P
Runner Q needs 10 × 32 = 320 seconds per kilometer; Runner P needs 300 seconds, so P covers a kilometer sooner.

202. HSPT-M16-028 - Answer B: 31.5 seconds
There are 63 intervals from the first click to the sixty-fourth; each lasts 60/120 = 0.5 second, giving 31.5 seconds.

203. HSPT-M16-029 - Answer C: 375 grams
Both readings contain the same added error, so subtracting cancels it as well as the container: 515 − 140 = 375 grams.

204. HSPT-M16-030 - Answer D: Should sensible travelers support the clearly superior new timetable?
Sensible and clearly superior praise one answer in advance, pressuring respondents toward agreement; the other questions do not attach approval to a particular response.

205. HSPT-M16-031 - Answer D: The direction of the association between the two variables
The separate coordinate lists retain the same values, means, and ranges, but larger first coordinates now accompany smaller rather than larger second coordinates.

206. HSPT-M16-032 - Answer A: 3/8
A works with probability 1/2. Among B and C, at least one works with probability 3/4, giving (1/2)(3/4) = 3/8.

207. HSPT-M16-033 - Answer A: 1/8 of a turn
Three advances total 9/8 of a turn. Subtracting one complete turn leaves 9/8 − 8/8 = 1/8 of a turn.

208. HSPT-M16-034 - Answer B: Infinitely many
The average of any two distinct rational numbers lies between them and is rational; repeatedly taking such averages produces more numbers in the interval.

209. HSPT-M16-035 - Answer C: 16 2/3%
Use a = 7k and b = 5k. The difference is 2k and the sum is 12k, so the requested fraction is 1/6.

210. HSPT-M16-036 - Answer A: 3
An odd factor count identifies a perfect square. A number that is both a square and a cube is a sixth power: 1, 64, or 729 here.

211. HSPT-M16-037 - Answer C: No real values
An absolute value is always nonnegative, so |x + 6| cannot equal the negative number −2 for any real input.

212. HSPT-M16-038 - Answer B: E = x for every real x except −2.
Factoring the numerator gives x(x + 2), which cancels to x only when x + 2 is nonzero; the excluded input remains −2.

213. HSPT-M16-039 - Answer C: 36/(x − 2)
Only x − 2 trays receive stones, so dividing the fixed total 36 by that occupied-tray count gives 36/(x − 2).

214. HSPT-M16-040 - Answer D: 21
Moving an input backward by 3 increases its output by 4. From 2 to −7 requires three backward moves, so f(−7) = 9 + 12 = 21.

215. HSPT-M16-041 - Answer D: (5, 5)
The move from A to D is 2 units right and 3 up. Applying the same move from B gives C = (5, 5).

216. HSPT-M16-042 - Answer C: 6 centimeters
The arc is 150/360 = 5/12 of the circumference. Thus the full circumference is 12π, and dividing by 2π gives radius 6.

217. HSPT-M16-043 - Answer A: 4 + 2√2 units
The outer stretched boundary has four segments of length 1 and two diagonal segments spanning 1-by-1 gaps, each of length √2.

218. HSPT-M16-044 - Answer B: 8 minutes 10 seconds
B finishes at 7:00, while A finishes at 7:30. Assembly therefore starts at 7:30 and ends 40 seconds later, at 8:10.

219. HSPT-M16-045 - Answer B: 2.00 meters
The washed width must be 180 + 2 + 2 = 184 centimeters. That is 92% of the prewash width, so 184/0.92 = 200 centimeters.

220. HSPT-M16-046 - Answer A: 2
The original range is 5 and median is 3. Adding 2 preserves endpoints 2 and 7 but makes the five-value median 2.

221. HSPT-M16-047 - Answer D: The mean stays 10, but the median may change.
The sum drops from 50 to 40 across four measurements, preserving mean 10. The two remaining middle values need not average 10.

222. HSPT-M16-048 - Answer D: 1/10
Only alternating triples ACE and BDF work. There are 6 × 5 × 4 ÷ 6 = 20 three-point sets, so the probability is 2/20 = 1/10.

223. HSPT-M16-049 - Answer C: 0.094
The corrected decimal is 0.500. Subtracting the original gives 0.500 − 0.406 = 0.094; the two digit changes have different place values.

224. HSPT-M16-050 - Answer A: 10
Since 200 = $2^3 \times 5^2$, n must contain factors 2 and 5 so its cube supplies both requirements. The least such n is 10.

225. HSPT-M16-051 - Answer A: 52%
Initially there are 100 grams of copper. Removal leaves 80 grams in 200 grams; adding pure copper gives 130 grams in 250 grams, or 52%.

226. HSPT-M16-052 - Answer A: x = 2 only
At x = 2, both sides equal 2. At x = −1, the square root equals 1 rather than −1, so that candidate is extraneous.

227. HSPT-M16-053 - Answer D: 27
There are n − 1 interior supports and 2 end supports, making n + 1 supports. Thus 2n + 1 = 55 and n = 27.

228. HSPT-M16-054 - Answer B: x = 3 and x = −1/3
Either 2x − 1 = x + 2, giving x = 3, or 2x − 1 = −x − 2, giving x = −1/3. Both satisfy the original.

229. HSPT-M16-055 - Answer C: 28 centimeters to the right
The trials give 2f − b = 17 and f − 2b = 1, so b = 5 and f = 11. Therefore 3f − b = 28.

230. HSPT-M16-056 - Answer A: A triangle
The plane meets one point on each of the three edges, and the three connecting segments bound a triangular face.

231. HSPT-M16-057 - Answer D: 5π + 4 units
Its outer arc is 3π and inner arc is 2π. Each of the two straight ends has length 6 − 4 = 2, giving 5π + 4.

232. HSPT-M16-058 - Answer B: 72 square meters
For width w, the other screened length is 24 − 2w. Area is w(24 − 2w) = 72 − 2(w − 6)², maximized at 72.

233. HSPT-M16-059 - Answer B: The surface area the material can cover
Square meters measure area, so coverage refers to a surface; boundary distance and thickness are lengths, while enclosed space is volume.

234. HSPT-M16-060 - Answer D: 6 grams
The stretched cord is 150 centimeters long, so 30 centimeters is one fifth of it. Uniform stretching preserves total mass, giving 30/5 = 6 grams.

235. HSPT-M16-061 - Answer B: 200 square meters
The original horizontal length is 3/1.5 = 2 centimeters. Actual dimensions are 2 × 5 = 10 meters and 4 × 5 = 20 meters, giving area 200.

236. HSPT-M16-062 - Answer C: The blank is missing information and cannot automatically be treated as zero.
A recorded zero reports that no damage was counted; a blank supplies no count, so replacing it with zero would add unsupported information.

237. HSPT-M16-063 - Answer D: The arithmetic mean
The lists 0, 1, 4, 5, 8 and 0, 3, 4, 7, 8 fit the summary but have means 3.6 and 4.4.

238. HSPT-M16-064 - Answer C: 1/2
The designated switch changes the majority exactly when the other two switches disagree. Those two disagree in 2 of their 4 equally likely settings.

Language

239. HSPT-L16-001 - Answer D: Will you, Captain, check the mooring before we leave?
Captain names the person being addressed inside the sentence, so commas separate it from the words on both sides.

240. HSPT-L16-002 - Answer A: the geese's footprints
Geese is an irregular plural without a final s; its possessive therefore adds apostrophe-s, while footprints remains an ordinary plural.

241. HSPT-L16-003 - Answer C: "I heard 'all clear,' but the gate remained closed," Ana reported.
Use single quotation marks inside double quotation marks; in standard US punctuation, both commas remain inside their respective closing quotation marks.

242. HSPT-L16-004 - Answer D: Professor Doran is a professor of engineering.
Professor is capitalized as a title directly before Doran; professor and engineering remain lowercase when used as ordinary descriptions.

243. HSPT-L16-005 - Answer B: Her schedule lists World History II, and she enjoys studying world history.
An official course title receives proper-name capitals, while a general reference to the subject world history remains lowercase.

244. HSPT-L16-006 - Answer A: giraffe
Giraffe is the standard spelling, with one r, two f's, and a final e after the doubled consonant.

245. HSPT-L16-007 - Answer D: iridescent
Iridescent uses one r and the ending escent; the alternatives double r, change a vowel, or omit c.

246. HSPT-L16-008 - Answer C: Only the front door was locked.
Placing only immediately before the front door limits which door was locked; the alternatives limit an action, a time, or the number of front doors.

247. HSPT-L16-009 - Answer C: inspect; to finish
Make takes an object followed by a bare verb, while persuade requires to before its verb, so inspect and to finish are correct.

248. HSPT-L16-010 - Answer B: Neither has the apprentice.
Neither preserves the negative meaning, and has matches the present-perfect statement and the singular subject apprentice.

249. HSPT-L16-011 - Answer D: dual
Dual means involving two functions or parts; duel names a contest, while duly and dull do not express the intended twofold role.

250. HSPT-L16-012 - Answer B: detract; distract
Detract from means diminish worth or appeal; distract means redirect attention, so the specified meanings require different verbs.

251. HSPT-L16-013 - Answer A: Putting a kite away properly protects its material and keeps its parts ready for the next use.
The paragraph explains drying and protecting the fabric as well as organizing parts for reassembly, so the topic sentence must cover both purposes.

252. HSPT-L16-014 - Answer C: Sentence 6
Sentence 6 concerns festival timing; the other sentences explain actions that protect the stored kite or keep its components available.

253. HSPT-L16-015 - Answer A: Note how the rods fit before removing them, then group matching pieces for later assembly.
The revision keeps observation before disassembly and grouping afterward; the alternatives reverse the useful sequence or introduce unsupported claims.

254. HSPT-L16-016 - Answer A: The invoice lists seventy-four bolts.
Compound numbers from twenty-one through ninety-nine take a hyphen when written as words, so seventy-four is correctly joined.

255. HSPT-L16-017 - Answer B: Yes, I checked the valve, but no, I did not open it.
Commas set off the introductory response words yes and no; another comma separates the complete clauses joined by but.

256. HSPT-L16-018 - Answer D: The container marked Co holds cobalt; the detector marked CO detects carbon monoxide.
Co uses one capital followed by lowercase o for cobalt; CO uses two capitals for carbon monoxide, so changing case changes the meaning of the notation.

257. HSPT-L16-019 - Answer B: NASA engineers tested the radar.
NASA retains its agency abbreviation's capitals; radar functions as an ordinary common noun, and engineers is also a common noun.

258. HSPT-L16-020 - Answer A: pomegranate
Pomegranate is the standard spelling; its middle portion is gran and its final syllable is written ate rather than ite or et.

259. HSPT-L16-021 - Answer C: labyrinth
Labyrinth is spelled with y after b and i after r; the alternatives substitute different vowels in those two positions.

260. HSPT-L16-022 - Answer B: desiccant
Desiccant has one s, two consecutive c's, and the ending ant; the alternatives alter one of those features.

261. HSPT-L16-023 - Answer C: Every spool has a numbered tag.
Every modifies a singular count noun, spool, which takes singular has; the alternatives use a plural noun or mismatched verb.

262. HSPT-L16-024 - Answer D: locking; to lock
Remember plus a gerund recalls an earlier action; remember plus a to-infinitive concerns carrying out a required future action.

263. HSPT-L16-025 - Answer D: The sail needs to be mended.
Needs mending gives the sail the role of receiving repair; needs to be mended preserves that role with a passive infinitive.

264. HSPT-L16-026 - Answer A: The new statute establishes the maximum permitted load.
A rule establishing a legal maximum is a statute; a person's physical height is stature, not a written law.

265. HSPT-L16-027 - Answer D: The travelers waited all together under the awning.
All together means in one group or place; altogether means entirely or completely and does not express the intended gathering.

266. HSPT-L16-028 - Answer C: Nevertheless
Nevertheless contrasts the flexible ticket's useful refund feature with the qualification that it is not best for everyone; the other choices misstate that relationship.

267. HSPT-L16-029 - Answer B: Immediately after sentence 6
Sentence 6 introduces the uncertain appointment, and the added sentence explains that uncertainty before sentence 7 describes the financial protection it makes relevant.

268. HSPT-L16-030 - Answer A: The better purchase depends on how likely the traveler is to need the protection that the higher price buys.
The two traveler examples support matching the refund option to the risk of changed plans, rather than prescribing one ticket for everyone.

269. HSPT-L16-031 - Answer B: "I'll take it." Nessa lifted the parcel.
Nessa lifted the parcel describes an action, not a way of speaking, so the quotation ends with a period before that separate sentence.

270. HSPT-L16-032 - Answer B: The carpenter restored a small round oak table.
These cumulative modifiers build from material to shape to size; they do not take the separating commas used between coordinate adjectives.

271. HSPT-L16-033 - Answer D: The choices were to rent a van, which offered more room; to repair the cart, which needed new wheels; or to postpone the delivery, which required everyone's consent.
Each semicolon belongs after a complete option, including its which-clause; placing it before which or inside that clause splits an option incorrectly.

272. HSPT-L16-034 - Answer A: October; Canadian
October is a month name and Canadian is a proper adjective; village and border are ordinary common nouns and remain lowercase.

273. HSPT-L16-035 - Answer A: The work involves (a) cleaning the joints, (b) replacing the seals, and (c) checking the pressure.
The labels do not create sentence boundaries; cleaning, replacing, and checking remain common words within the same continuous sentence.

274. HSPT-L16-036 - Answer C: skiing
Skiing keeps the i in ski and adds ing, producing two consecutive i's; it does not drop one or insert a third.

275. HSPT-L16-037 - Answer D: mannequin
Mannequin has two n's after ma and the ending equin; the alternatives omit a consonant or alter that final letter sequence.

276. HSPT-L16-038 - Answer B: some
Information is a noncount noun in this context, so some can introduce it; a, an, and several require a countable noun.

277. HSPT-L16-039 - Answer C: At 10:05, Rosa was copying the chart.
The past progressive was copying describes an activity already underway at the specified past moment and agrees with the singular subject Rosa.

278. HSPT-L16-040 - Answer A: The keeper told Ari not to disturb the seals.
A reported command after told plus an object uses a to-infinitive; placing not before it correctly reports the negative instruction.

279. HSPT-L16-041 - Answer C: rationale; rational
Rationale is a noun meaning the reasons behind a choice; rational is an adjective meaning reasonable, so the two positions require different forms.

280. HSPT-L16-042 - Answer C: palate; palette; pallet
Palate concerns taste, palette holds an artist's colors, and pallet supports freight; the similar spellings refer to three distinct things.

281. HSPT-L16-043 - Answer D: regimen; regiment
Regimen means an organized routine or plan; regiment names a military unit, so the intended practice and membership contexts require different nouns.

282. HSPT-L16-044 - Answer B: 1, 3, 5, 2, 4, 6
Sentence 3 supplies the original claim, 5 explains its condition, 2 removes it, and 4 shows the result before 6 proposes a fair remedy.

283. HSPT-L16-045 - Answer A: Preserving the Limits of a Quoted Claim
The passage explains how retaining or fairly explaining a condition preserves a quotation's meaning; it does not reject all short quotations or teach sorting.

284. HSPT-L16-046 - Answer C: The courier, a cyclist carrying a blue satchel, waited outside.
The explanatory appositive a cyclist carrying a blue satchel interrupts the main sentence, so a comma marks each of its two boundaries.

285. HSPT-L16-047 - Answer A: The notice says the north gate opens "at dawn."
The exact phrase at dawn fits directly into the reported clause without an introducing comma; its closing quotation mark follows the final period.

286. HSPT-L16-048 - Answer C: The switch was off; the screen stayed dark.
The semicolon continues the same sentence, so the common word the after it remains lowercase; only the sentence's opening word requires a capital.

287. HSPT-L16-049 - Answer C: The central processing unit (CPU) coordinates the calculations.
The ordinary expanded term stays lowercase, while its conventional abbreviation is CPU; uppercase initials do not make the full descriptive phrase a proper name.

288. HSPT-L16-050 - Answer B: cantaloupe
Cantaloupe includes a after t and the final sequence oupe; the other spellings change that vowel or omit or replace letters in the ending.

289. HSPT-L16-051 - Answer D: crescendo
Crescendo has one s followed by c and ends endo; the alternatives double s, omit c, or insert an extra h.

290. HSPT-L16-052 - Answer A: chrysanthemum
Chrysanthemum begins chrys, contains anthem, and ends um; the alternatives omit h, replace a vowel, or change the final consonant.

291. HSPT-L16-053 - Answer B: surprised the crew
The what-clause serves as the subject; surprised the crew supplies the main predicate, whereas the other completions add only modifiers.

292. HSPT-L16-054 - Answer C: They must have taken the early boat; their berth is empty.
Must have taken expresses a strong conclusion about an earlier completed action; the alternatives express present obligation or an action occurring now.

293. HSPT-L16-055 - Answer A: stay; arrived
Would rather takes a bare verb for the speaker's own action; a separate subject takes a past form to express a present or future preference.

294. HSPT-L16-056 - Answer D: climatic; climactic
Climatic relates to climate, while climactic describes a climax or point of greatest intensity; the map and novel therefore require different adjectives.

295. HSPT-L16-057 - Answer B: prophesy; prophecy
Prophesy is the verb meaning to predict; prophecy is the noun naming a prediction, so will and the require those respective forms.

296. HSPT-L16-058 - Answer A: incidence; incidents
Incidence denotes the rate or frequency of occurrence; incidents are individual events, so the specified rate and event meanings require different nouns.

297. HSPT-L16-059 - Answer D: In a test of the grayscale copy, readers distinguished the solid and dashed routes even where their gray tones matched.
The test directly connects line patterns with distinguishing routes when color is unavailable; the other details concern appearance, production speed, or design inspiration.

298. HSPT-L16-060 - Answer D: Distinct line patterns will help readers follow each route in a copy that has no usable color.
The revision names line patterns, readers, routes, and the copying condition; the alternatives remain vague or add an unsupported rejection of all color printing.